Lab Assignment

International Cybersecurity and Digital Forensics Academy

"Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training"

1. Purpose of the Lab Assignment

The purpose of this lab is to investigate a Windows memory image for evidence of a suspected unauthorized file transfer. It involves using forensic tools to examine computer processes, network connections, command history, and user activities. The investigation also focuses on identifying USB devices, drive mappings, and File Explorer records. The aim is to understand how these pieces of evidence may be connected to the suspected transfer.

2. Lab Report Structure

The report begins with the case reference and student identification, followed by the executive summary. It explains how the memory image was collected, preserved, and checked using its file details and SHA-256 hash. The findings section presents the results of the Volatility analysis, including processes, network records, command history, USB devices, and ShellBags.

2.1. Title Page

- Lab Title: Case Study 3: Investigating Memory Evidence
- Student Name: Olanrewaju Wale Mustapha Oladimeji.
- Student ID: 2025/FWSD/11320.
- Course Name: SBT-DF204: Computer Forensic Case Study
- Instructor Name: Aminu Idris
- Date of Submission: 18/10/2026
- Version:

2.2. Executive Summary

This investigation examines a Windows memory image to determine whether available evidence supports a suspected unauthorized file transfer. Volatility 2 is used to identify the operating system profile and examine relevant processes, network connections, and command history.

The investigation also checks registry records for USB devices, drive mappings, user profiles, and File Explorer activity. The findings are compared to establish whether the different records support a possible sequence of events.

2.3. Lab Objectives

The first objective is to preserve the memory image and verify its integrity using file details and a SHA-256 hash. The second objective is to identify the appropriate Windows memory profile and examine relevant processes and network activity. The third objective is to recover and assess command-history records that may relate to file copying or transfer attempts. The fourth objective is to investigate USB devices, registry records, user profiles, and ShellBags for evidence that may support the suspected activity. The final objective is to create an evidence log and UTC timeline and provide a careful conclusion based on the available findings.

2.4. Tools and Resources Used

Kali Linux is used as the forensic workstation for carrying out the memory investigation.

Volatility is the main forensic tool used to examine the memory image and recover relevant system artefacts.

Python 2 is used to run Volatility 2 in accordance with the supplied laboratory instructions.

Linux terminal commands, including stat, sha256sum, cp, and find, are used to manage files, record file details, locate the memory image, and verify evidence integrity.

The supplied memdumpWin7.mem memory image, saved command outputs, screenshots, evidence log, and timeline are used to document and support the investigation.

2.5. Methodology

This individual case study asks you to examine a Windows 7 memory image and reconstruct an evidence-based account of a suspected illegal file transfer. Use Volatility 2 to identify relevant processes, network artefacts, command history, removable-media records, user-profile evidence, and File Explorer traces. Your report must clearly separate direct observations from inferences about a transfer or the person who used the workstation.

Case scenario

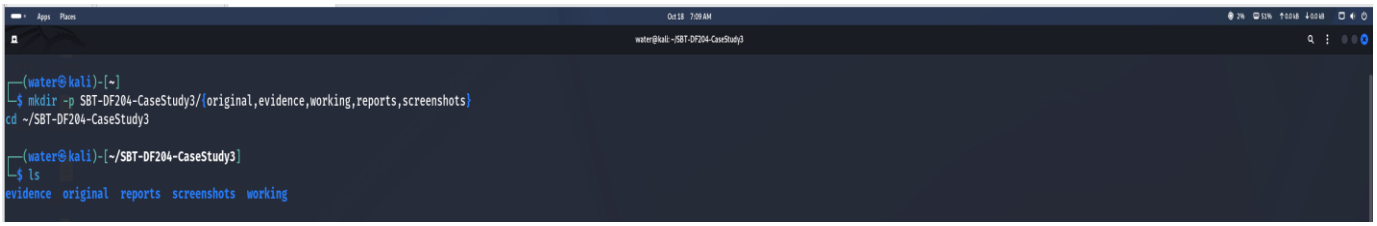
Investigators obtained a Windows 10 memory image from a workstation after a confidential document may have been transferred without authorization. The inquiry is whether browser activity, a command prompt, a removable device, and a user profile form a defensible

sequence. This is an offline examination of the supplied material. It does not authorize contact with any address, host, account, file-sharing service, or person that may appear in the image.

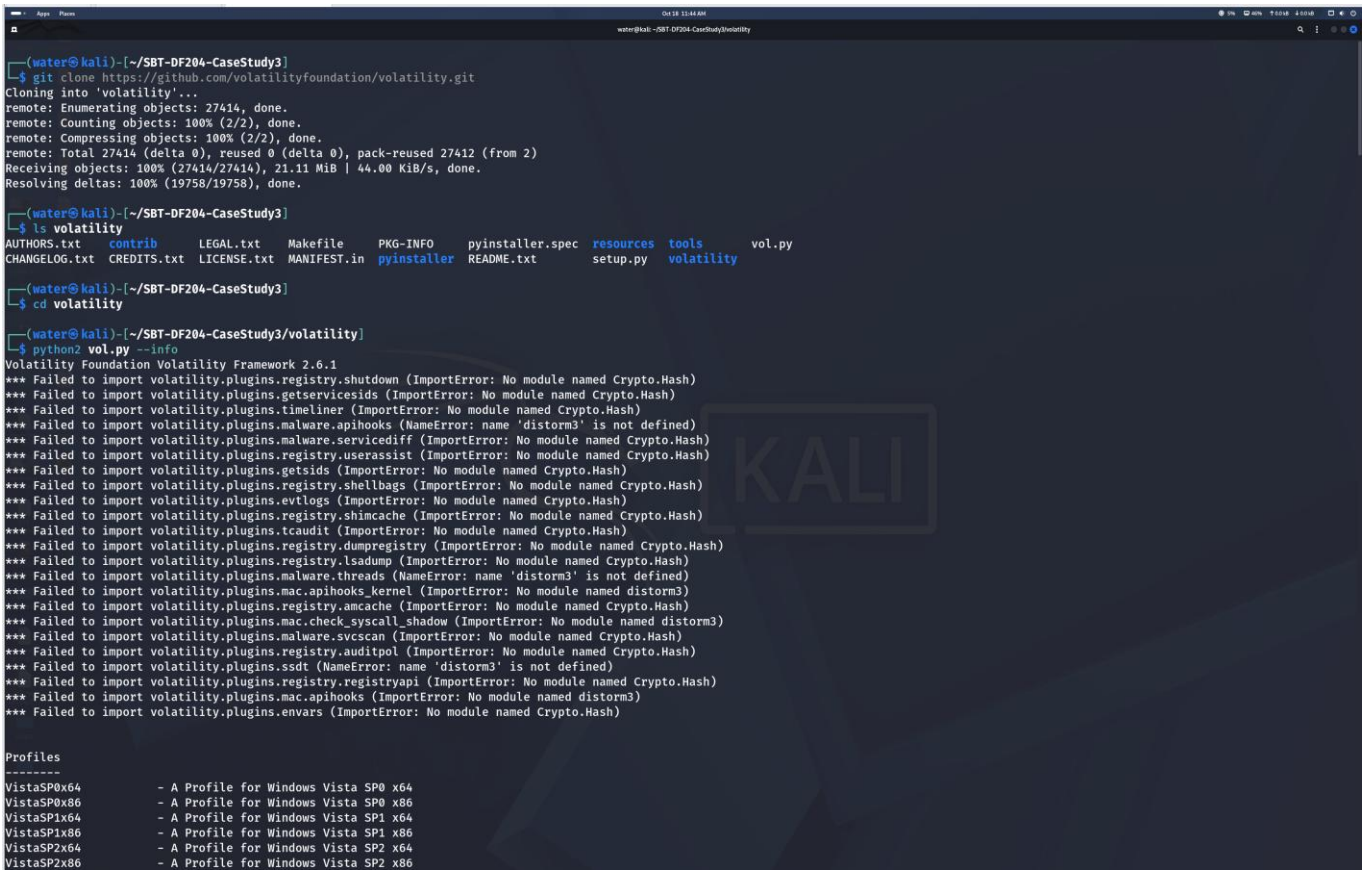
Case materials

The supplied case file is a Windows 7 memory image named memdumpWin7.mem. The teaching deck demonstrates Volatility 2 methods. In the ICDFA Kali lab, start Volatility 2 with python2 vol.py. Create a working copy before analysis. Record the source, original file name, file size, SHA-256 value, and tool version. Select and justify the memory profile from the actual image output instead of assuming one from a screenshot.

Creating a working Folder for Case Study 3



Installing Volatility



Downloading memdumpWin7.mem



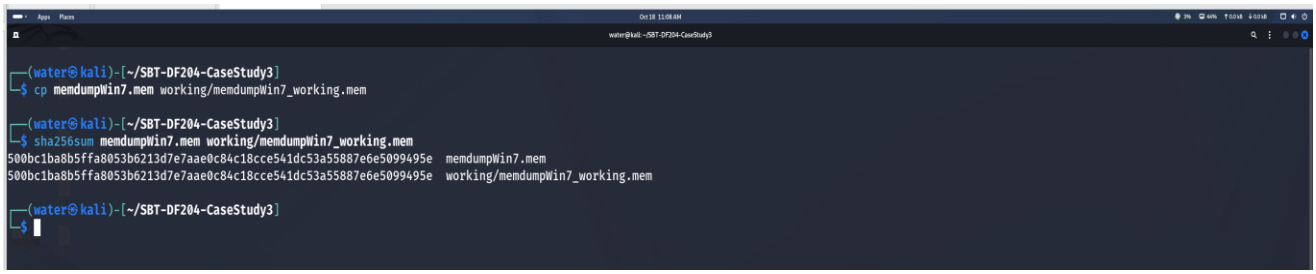
Confirming the file size and Name



Checking the SHA256sum



Creating a working copy of memdumpWin7.mem/sha256sum

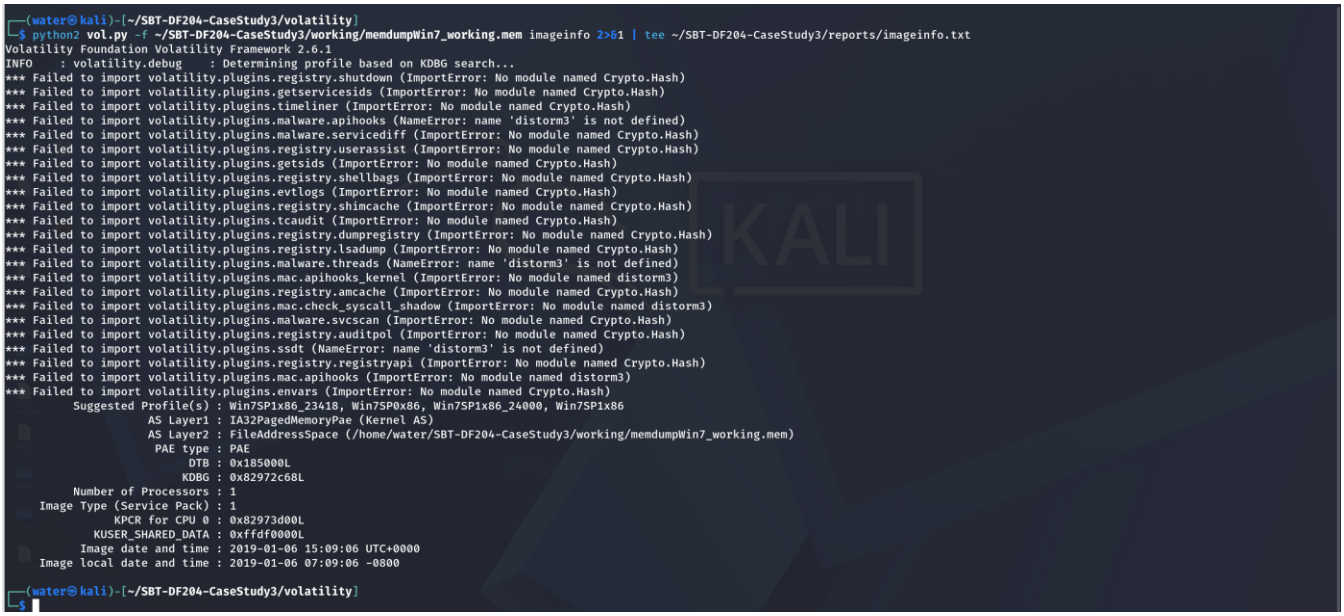


sha256sum memdumpWin7.mem working/memdumpWin7_working.mem
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e
memdumpWin7.mem

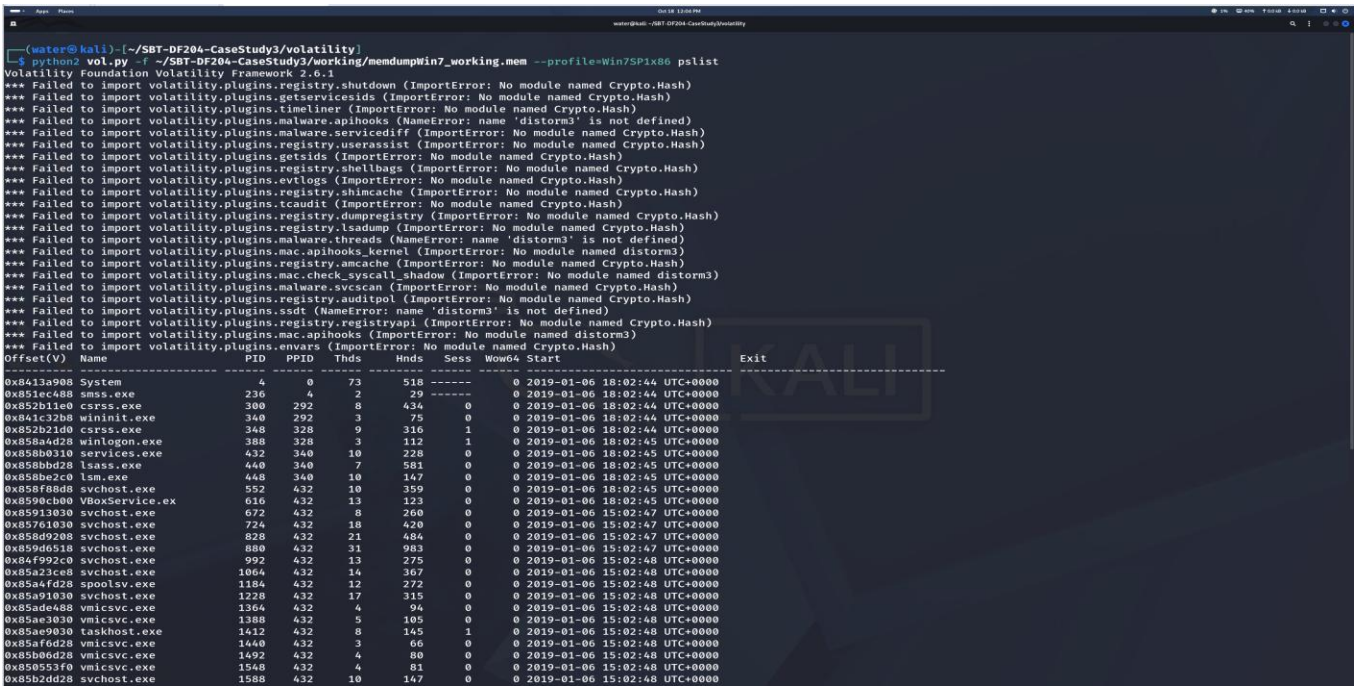
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e
working/memdumpWin7_working.mem

Volatility 2 examination starting points

Identify the image profile and capture time



List active and residual processes/List active processes



```
0x85b2dd28 svchost.exe            1588    432     10      147      0      0 2019-01-06 15:02:48 UTC+0000
0x859dc030 cygrunsrv.exe          1680    432      6      100      0      0 2019-01-06 15:02:48 UTC+0000
0x859a0d28 wlms.exe               1752    432      4       45      0      0 2019-01-06 15:02:48 UTC+0000
0x859bab00 taskeng.exe            1776    880      6       85      0      0 2019-01-06 15:02:48 UTC+0000
0x8500bd00 cygrunsrv.exe          1984   1680      0 --------      0      0 2019-01-06 15:02:49 UTC+0000   2019-01-06 15:02:49 UTC+0000
0x84202030 conhost.exe            2000    300      2       33      0      0 2019-01-06 15:02:49 UTC+0000
0x84203998 sshd.exe               2016   1984      4      100      0      0 2019-01-06 15:02:49 UTC+0000
0x8501cce8 GoogleUpdate.e          192   1776      4      109      0      0 2019-01-06 15:02:49 UTC+0000
0x85b11398 sppsvc.exe             1036    432      4      143      0      0 2019-01-06 15:02:50 UTC+0000
0x85a07030 svchost.exe            2072    432      5       91      0      0 2019-01-06 15:02:50 UTC+0000
0x857ce6d0 rundll32.exe           2320    552      4       85      1      0 2019-01-06 15:02:52 UTC+0000
0x84fddd28 dwm.exe                2512    828      3       70      1      0 2019-01-06 15:02:54 UTC+0000
0x85bba860 explorer.exe           2524   2500     32      952      1      0 2019-01-06 15:02:54 UTC+0000
0x85c7f030 VBoxTray.exe           2632   2524     13      140      1      0 2019-01-06 15:02:55 UTC+0000
0x85cb9820 SearchIndexer.         2960    432     13      607      0      0 2019-01-06 15:03:01 UTC+0000
0x85e287f0 WUDFHost.exe           3300    828      8      205      0      0 2019-01-06 15:03:07 UTC+0000
0x84fa2ad0 msiexec.exe            3748    432      4      174      0      0 2019-01-06 15:03:34 UTC+0000
0x841eb658 WmiPrvSE.exe           4032    552      5      114      0      0 2019-01-06 15:03:54 UTC+0000
0x8583a030 chrome.exe             2388   2524     33      819      1      0 2019-01-06 15:04:30 UTC+0000
0x851e8520 chrome.exe             1280   2388      7       82      1      0 2019-01-06 15:04:30 UTC+0000
0x850316b8 chrome.exe             2380   2388      2       57      1      0 2019-01-06 15:04:30 UTC+0000
0x852845c8 svchost.exe            2416    432     10      139      0      0 2019-01-06 15:04:30 UTC+0000
0x842b4d28 chrome.exe             2912   2388      9      159      1      0 2019-01-06 15:04:32 UTC+0000
0x858be710 svchost.exe            3784    432     14      370      0      0 2019-01-06 15:04:51 UTC+0000
0x8431fb08 cmd.exe                3996   2524      1       22      1      0 2019-01-06 15:06:11 UTC+0000
0x842ddd28 conhost.exe            3988    348      2       52      1      0 2019-01-06 15:06:11 UTC+0000
0x842eb6b0 chrome.exe              976   2388     14      198      1      0 2019-01-06 15:06:23 UTC+0000
0x84312030 FTK Imager.exe         2596   2524     14      358      1      0 2019-01-06 15:07:15 UTC+0000
0x850d5a38 dllhost.exe            2228    552      6 170...38      1      0 2019-01-06 15:09:07 UTC+0000

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$
```

Residual Process

```
┌──(water㉿kali)-[~]
└─$ cd ~/SBT-DF204-CaseStudy3/volatility

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$ python2 ~/SBT-DF204-CaseStudy3/volatility/vol.py \
-f ~/SBT-DF204-CaseStudy3/working/memdumpWin7_working.mem \
--profile=Win7SP1x86 psscan \
2>&1 | tee ~/SBT-DF204-CaseStudy3/reports/psscan.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Offset(P)          Name                PID   PPID PDB        Time created                   Time exited
------------------ ---------------- ------ ------ ---------- ------------------------------ ------------------------------
```

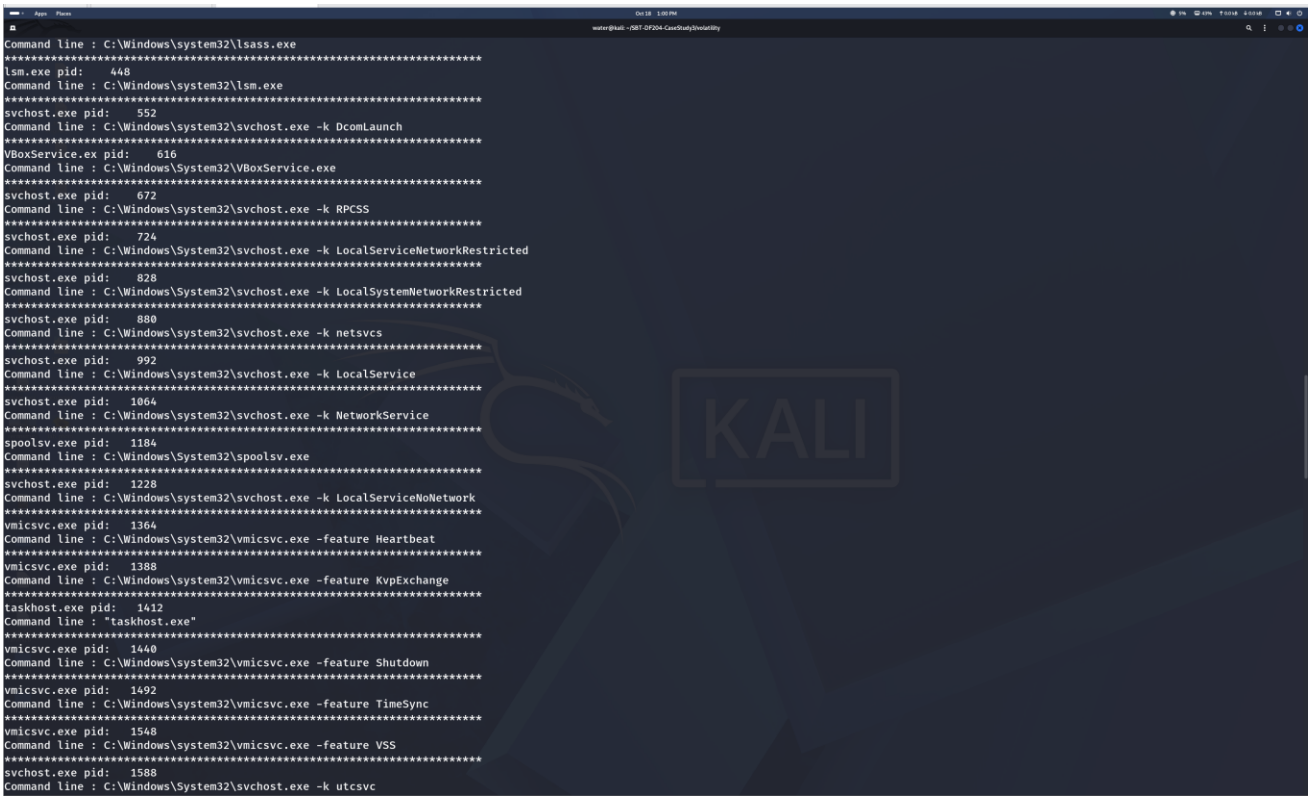
Checking Process Hierarchy

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$ python2 ~/SBT-DF204-CaseStudy3/volatility/vol.py \
-f ~/SBT-DF204-CaseStudy3/working/memdumpWin7_working.mem \
--profile=Win7SP1x86 pstree \
2>&1 | tee ~/SBT-DF204-CaseStudy3/reports/pstree.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Name                                                  Pid   PPid   Thds   Hnds Time
----------------------------------------------- ------ ------ ------ ------ ----
 0x85bba860:explorer.exe                             2524   2500     32    952 2019-01-06 15:02:54 UTC+0000
. 0x8431fb08:cmd.exe                                 3996   2524      1     22 2019-01-06 15:06:11 UTC+0000
. 0x85c7f030:VBoxTray.exe                            2632   2524     13    140 2019-01-06 15:02:55 UTC+0000
. 0x8583a030:chrome.exe                              2388   2524     33    819 2019-01-06 15:04:30 UTC+0000
.. 0x851e8520:chrome.exe                             1280   2388      7     82 2019-01-06 15:04:30 UTC+0000
.. 0x842eb6b0:chrome.exe                              976   2388     14    198 2019-01-06 15:06:23 UTC+0000
.. 0x850316b8:chrome.exe                             2380   2388      2     57 2019-01-06 15:04:30 UTC+0000
.. 0x842b4d28:chrome.exe                             2912   2388      9    159 2019-01-06 15:04:32 UTC+0000
. 0x84312030:FTK Imager.exe                          2596   2524     14    358 2019-01-06 15:07:15 UTC+0000
 0x852b11e0:csrss.exe                                 300    292      8    434 2019-01-06 18:02:44 UTC+0000
. 0x84202030:conhost.exe                             2000    300      2     33 2019-01-06 15:02:49 UTC+0000
 0x841c32b8:wininit.exe                               340    292      3     75 2019-01-06 18:02:44 UTC+0000
. 0x858b0310:services.exe                             432    340     10    228 2019-01-06 18:02:45 UTC+0000
.. 0x85ae9030:taskhost.exe                           1412    432      8    145 2019-01-06 15:02:48 UTC+0000
.. 0x85ae3030:vmicsvc.exe                            1388    432      5    105 2019-01-06 15:02:48 UTC+0000
.. 0x85b06d28:vmicsvc.exe                            1492    432      4     80 2019-01-06 15:02:48 UTC+0000
.. 0x85b11398:sppsvc.exe                             1036    432      4    143 2019-01-06 15:02:50 UTC+0000
.. 0x859dc030:cygrunsrv.exe                          1680    432      6    100 2019-01-06 15:02:48 UTC+0000
... 0x8500bd00:cygrunsrv.exe                         1984   1680      0 ------ 2019-01-06 15:02:49 UTC+0000
```

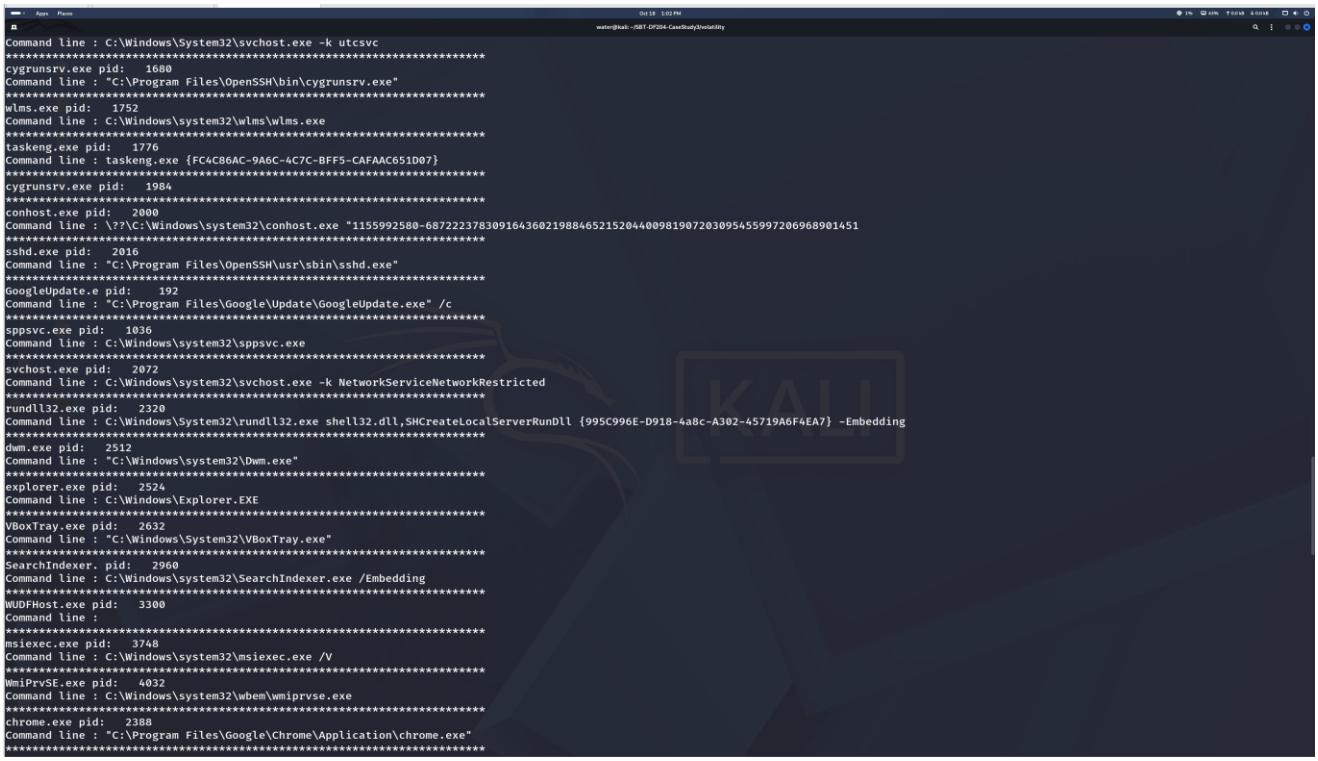
```
... 0x8500bd00:cygrunsrv.exe                    1984   1680      0 ------ 2019-01-06 15:02:49 UTC+0000
.... 0x84203998:sshd.exe                        2016   1984      4      100 2019-01-06 15:02:49 UTC+0000
.. 0x85a07030:svchost.exe                       2072    432      5       91 2019-01-06 15:02:50 UTC+0000
.. 0x85913030:svchost.exe                        672    432      8      260 2019-01-06 15:02:47 UTC+0000
.. 0x84fa2ad0:msiexec.exe                       3748    432      4      174 2019-01-06 15:03:34 UTC+0000
.. 0x858f88d8:svchost.exe                        552    432     10      359 2019-01-06 18:02:45 UTC+0000
... 0x850d5a38:dllhost.exe                      2228    552      6 17...8 2019-01-06 15:09:07 UTC+0000
... 0x841eb658:WmiPrvSE.exe                     4032    552      5      114 2019-01-06 15:03:54 UTC+0000
... 0x857ce6d0:rundll32.exe                     2320    552      4       85 2019-01-06 15:02:52 UTC+0000
.. 0x858be710:svchost.exe                       3784    432     14      370 2019-01-06 15:04:51 UTC+0000
.. 0x85b2dd28:svchost.exe                       1588    432     10      147 2019-01-06 15:02:48 UTC+0000
.. 0x858d9208:svchost.exe                        828    432     21      484 2019-01-06 15:02:47 UTC+0000
... 0x84fddd28:dwm.exe                          2512    828      3       70 2019-01-06 15:02:54 UTC+0000
... 0x85e287f0:WUDFHost.exe                     3300    828      8      205 2019-01-06 15:03:07 UTC+0000
.. 0x85a4fd28:spoolsv.exe                       1184    432     12      272 2019-01-06 15:02:48 UTC+0000
.. 0x85cb9820:SearchIndexer.                    2960    432     13      607 2019-01-06 15:03:01 UTC+0000
.. 0x850553f0:vmicsvc.exe                       1548    432      4       81 2019-01-06 15:02:48 UTC+0000
.. 0x85a91030:svchost.exe                       1228    432     17      315 2019-01-06 15:02:48 UTC+0000
.. 0x85761030:svchost.exe                        724    432     18      420 2019-01-06 15:02:47 UTC+0000
.. 0x859a0d28:wlms.exe                          1752    432      4       45 2019-01-06 15:02:48 UTC+0000
.. 0x84f992c0:svchost.exe                        992    432     13      275 2019-01-06 15:02:48 UTC+0000
.. 0x8590cb00:VBoxService.ex                     616    432     13      123 2019-01-06 18:02:45 UTC+0000
.. 0x859d6518:svchost.exe                        880    432     31      983 2019-01-06 15:02:47 UTC+0000
... 0x859bab00:taskeng.exe                      1776    880      6       85 2019-01-06 15:02:48 UTC+0000
.... 0x8501cce8:GoogleUpdate.e                   192   1776      4      109 2019-01-06 15:02:49 UTC+0000
.. 0x85a23ce8:svchost.exe                       1064    432     14      367 2019-01-06 15:02:48 UTC+0000
.. 0x85af6d28:vmicsvc.exe                       1440    432      3       66 2019-01-06 15:02:48 UTC+0000
.. 0x852845c8:svchost.exe                       2416    432     10      139 2019-01-06 15:04:30 UTC+0000
.. 0x85ade488:vmicsvc.exe                       1364    432      4       94 2019-01-06 15:02:48 UTC+0000
. 0x858bbd28:lsass.exe                           440    340      7      581 2019-01-06 18:02:45 UTC+0000
. 0x858be2c0:lsm.exe                             448    340     10      147 2019-01-06 18:02:45 UTC+0000
 0x858a4d28:winlogon.exe                         388    328      3      112 2019-01-06 18:02:45 UTC+0000
 0x852b21d0:csrss.exe                            348    328      9      316 2019-01-06 18:02:44 UTC+0000
. 0x842ddd28:conhost.exe                        3988    348      2       52 2019-01-06 15:06:11 UTC+0000
 0x8413a908:System                                 4      0     73      518 2019-01-06 18:02:44 UTC+0000
. 0x851ec488:smss.exe                            236      4      2       29 2019-01-06 18:02:44 UTC+0000
```

Checking the Process Command-line

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$ python2 ~/SBT-DF204-CaseStudy3/volatility/vol.py \
-f ~/SBT-DF204-CaseStudy3/working/memdumpWin7_working.mem \
--profile=Win7SP1x86 cmdline \
2>&1 | tee ~/SBT-DF204-CaseStudy3/reports/cmdline.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
**************************************************
System pid:      4
**************************************************
smss.exe pid:    236
Command line : \SystemRoot\System32\smss.exe
**************************************************
csrss.exe pid:    300
Command line : %SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,12288,512 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServerDllInitializat
ion,3 ServerDll=winsrv:ConServerDllInitialization,2 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16
**************************************************
wininit.exe pid:    340
Command line : wininit.exe
**************************************************
csrss.exe pid:    348
Command line : %SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,12288,512 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServerDllInitializat
ion,3 ServerDll=winsrv:ConServerDllInitialization,2 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16
**************************************************
winlogon.exe pid:    388
Command line : winlogon.exe
**************************************************
services.exe pid:    432
Command line : C:\Windows\system32\services.exe
**************************************************
lsass.exe pid:    440
Command line : C:\Windows\system32\lsass.exe
```



```
Command line : C:\Windows\system32\lsass.exe
**********************************************************************
lsm.exe pid:    448
Command line : C:\Windows\system32\lsm.exe
**********************************************************************
svchost.exe pid:    552
Command line : C:\Windows\system32\svchost.exe -k DcomLaunch
**********************************************************************
VBoxService.ex pid:    616
Command line : C:\Windows\System32\VBoxService.exe
**********************************************************************
svchost.exe pid:    672
Command line : C:\Windows\system32\svchost.exe -k RPCSS
**********************************************************************
svchost.exe pid:    724
Command line : C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted
**********************************************************************
svchost.exe pid:    828
Command line : C:\Windows\System32\svchost.exe -k LocalSystemNetworkRestricted
**********************************************************************
svchost.exe pid:    880
Command line : C:\Windows\system32\svchost.exe -k netsvcs
**********************************************************************
svchost.exe pid:    992
Command line : C:\Windows\system32\svchost.exe -k LocalService
**********************************************************************
svchost.exe pid:   1064
Command line : C:\Windows\system32\svchost.exe -k NetworkService
**********************************************************************
spoolsv.exe pid:   1184
Command line : C:\Windows\System32\spoolsv.exe
**********************************************************************
svchost.exe pid:   1228
Command line : C:\Windows\system32\svchost.exe -k LocalServiceNoNetwork
**********************************************************************
vmicsvc.exe pid:   1364
Command line : C:\Windows\system32\vmicsvc.exe -feature Heartbeat
**********************************************************************
vmicsvc.exe pid:   1388
Command line : C:\Windows\system32\vmicsvc.exe -feature KvpExchange
**********************************************************************
taskhost.exe pid:   1412
Command line : "taskhost.exe"
**********************************************************************
vmicsvc.exe pid:   1440
Command line : C:\Windows\system32\vmicsvc.exe -feature Shutdown
**********************************************************************
vmicsvc.exe pid:   1492
Command line : C:\Windows\system32\vmicsvc.exe -feature TimeSync
**********************************************************************
vmicsvc.exe pid:   1548
Command line : C:\Windows\system32\vmicsvc.exe -feature VSS
**********************************************************************
svchost.exe pid:   1588
Command line : C:\Windows\System32\svchost.exe -k utcsvc
```



```
Command line : C:\Windows\System32\svchost.exe -k utcsvc
**********************************************************************
cygrunsrv.exe pid:   1680
Command line : "C:\Program Files\OpenSSH\bin\cygrunsrv.exe"
**********************************************************************
wlms.exe pid:   1752
Command line : C:\Windows\system32\wlms\wlms.exe
**********************************************************************
taskeng.exe pid:   1776
Command line : taskeng.exe {FC4C86AC-9A6C-4C7C-BFF5-CAFAAC651D07}
**********************************************************************
cygrunsrv.exe pid:   1984
**********************************************************************
conhost.exe pid:   2000
Command line : \??\C:\Windows\system32\conhost.exe "1155992580-687222378309164360219884652152044009819072030954559972069689014 51"
**********************************************************************
sshd.exe pid:   2016
Command line : "C:\Program Files\OpenSSH\usr\sbin\sshd.exe"
**********************************************************************
GoogleUpdate.e pid:    192
Command line : "C:\Program Files\Google\Update\GoogleUpdate.exe" /c
**********************************************************************
sppsvc.exe pid:   1036
Command line : C:\Windows\system32\sppsvc.exe
**********************************************************************
svchost.exe pid:   2072
Command line : C:\Windows\system32\svchost.exe -k NetworkServiceNetworkRestricted
**********************************************************************
rundll32.exe pid:   2320
Command line : C:\Windows\System32\rundll32.exe shell32.dll,SHCreateLocalServerRunDll {995C996E-D918-4a8c-A302-45719A6F4EA7} -Embedding
**********************************************************************
dwm.exe pid:   2512
Command line : "C:\Windows\system32\Dwm.exe"
**********************************************************************
explorer.exe pid:   2524
Command line : C:\Windows\Explorer.EXE
**********************************************************************
VBoxTray.exe pid:   2632
Command line : "C:\Windows\System32\VBoxTray.exe"
**********************************************************************
SearchIndexer. pid:   2960
Command line : C:\Windows\system32\SearchIndexer.exe /Embedding
**********************************************************************
WUDFHost.exe pid:   3300
Command line :
**********************************************************************
msiexec.exe pid:   3748
Command line : C:\Windows\system32\msiexec.exe /V
**********************************************************************
WmiPrvSE.exe pid:   4032
Command line : C:\Windows\system32\wbem\wmiprvse.exe
**********************************************************************
chrome.exe pid:   2388
Command line : "C:\Program Files\Google\Chrome\Application\chrome.exe"
**********************************************************************
```

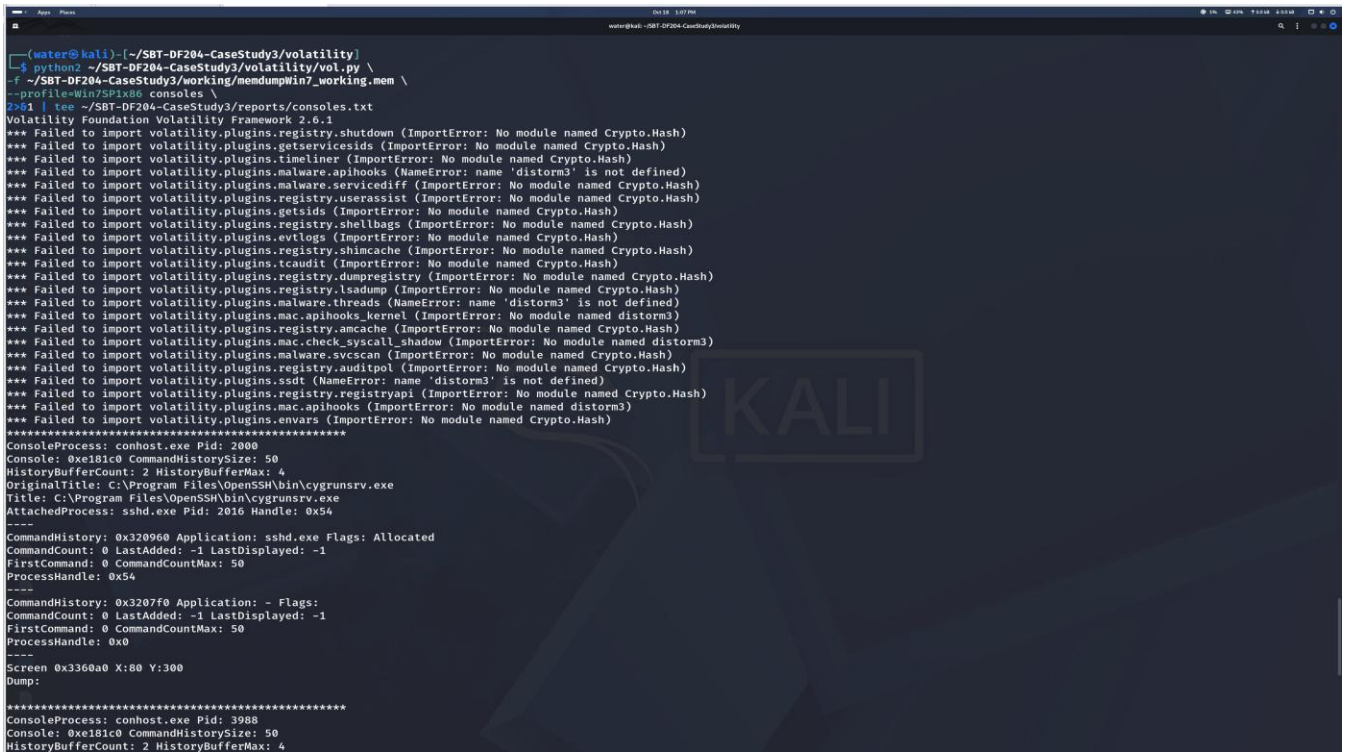
```
**************************************************************************
chrome.exe pid:   2388
Command line : "C:\Program Files\Google\Chrome\Application\chrome.exe"
**************************************************************************
chrome.exe pid:   1280
Command line :
**************************************************************************
chrome.exe pid:   2380
Command line : "C:\Program Files\Google\Chrome\Application\chrome.exe" --type=watcher --main-thread-id=1292 --on-initialized-event-handle=308 --parent-handle=312 /prefetch:6
**************************************************************************
svchost.exe pid:   2416
Command line : C:\Windows\system32\svchost.exe -k LocalServiceAndNoImpersonation
**************************************************************************
chrome.exe pid:   2912
Command line :
**************************************************************************
svchost.exe pid:   3784
Command line : C:\Windows\System32\svchost.exe -k secsvcs
**************************************************************************
cmd.exe pid:   3996
Command line : "C:\Windows\system32\cmd.exe"
**************************************************************************
conhost.exe pid:   3988
Command line : \??\C:\Windows\system32\conhost.exe "-164647720-2759156771236404584117504074 2-950872444-977965776-107363114-188091992
**************************************************************************
chrome.exe pid:    976
Command line : "C:\Program Files\Google\Chrome\Application\chrome.exe" --type=renderer --field-trial-handle=904,1792691977862110708,4827585219111034411,131072 --disable-gpu-compositing --service-
pipe-token=4706400379687013043 --lang=en-US --enable-offline-auto-reload --enable-offline-auto-reload-visible-only --device-scale-factor=1 --num-raster-threads=1 --service-request-channel-token=4
706400379687013043 --renderer-client-id=13 --mojo-platform-channel-handle=3460 /prefetch:1
**************************************************************************
FTK Imager.exe pid:   2596
Command line : "C:\Program Files\AccessData\FTK Imager\FTK Imager.exe"
**************************************************************************
dllhost.exe pid:   2228
Command line : C:\Windows\system32\DllHost.exe /Processid:{AB8902B4-09CA-4BB6-B78D-A8F59079A8D5}
```

Review network artefacts

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$ python2 ~/SBT-DF204-CaseStudy3/volatility/vol.py \
-f ~/SBT-DF204-CaseStudy3/working/memdumpWin7_working.mem \
--profile=Win7SP1x86 netscan \
2>&1 | tee ~/SBT-DF204-CaseStudy3/reports/netscan.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Offset(P)          Proto    Local Address                  Foreign Address        State            Pid      Owner          Created
0x2148bbb0         TCPv4    0.0.0.0:49155                  0.0.0.0:0              LISTENING        432      services.exe
0x269ff708         TCPv4    0.0.0.0:49156                  0.0.0.0:0              LISTENING        440      lsass.exe
0x269ff708         TCPv6    :::49156                       :::0                   LISTENING        440      lsass.exe
0x3e081f50         UDPv4    192.168.56.8:56647             *:*                                     2388     chrome.exe     2019-01-06 15:08:32 UTC+0000
0x3e2863d8         UDPv4    0.0.0.0:0                      *:*                                     616      VBoxService.ex 2019-01-06 15:08:35 UTC+0000
0x3e2a2480         UDPv6    fe80::1571:7b34:b2b1:bbe4:1900 *:*                                     2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e46edf0         UDPv4    192.168.56.8:137               *:*                                     4        System         2019-01-06 15:02:48 UTC+0000
0x3e46ef50         UDPv4    192.168.56.8:138               *:*                                     4        System         2019-01-06 15:02:48 UTC+0000
0x3e5c0f50         UDPv4    0.0.0.0:0                      *:*                                     2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c1d80         UDPv4    0.0.0.0:0                      *:*                                     2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c1d80         UDPv6    :::0                           *:*                                     2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c5910         UDPv4    0.0.0.0:5355                   *:*                                     1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c7590         UDPv4    0.0.0.0:5355                   *:*                                     1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c7590         UDPv6    :::5355                        *:*                                     1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e6074e0         UDPv6    ::1:51122                      *:*                                     2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e7f2828         UDPv4    0.0.0.0:0                      *:*                                     1064     svchost.exe    2019-01-06 15:02:49 UTC+0000
0x3e7f2828         UDPv6    :::0                           *:*                                     1064     svchost.exe    2019-01-06 15:02:49 UTC+0000
0x3e446008         TCPv4    0.0.0.0:49154                  0.0.0.0:0              LISTENING        880      svchost.exe
0x3e446d98         TCPv4    0.0.0.0:49154                  0.0.0.0:0              LISTENING        880      svchost.exe
0x3e446d98         TCPv6    :::49154                       :::0                   LISTENING        880      svchost.exe
0x3e46e0c0         TCPv4    192.168.56.8:139               0.0.0.0:0              LISTENING        4        System
0x3e671988         TCPv4    0.0.0.0:445                    0.0.0.0:0              LISTENING        4        System
0x3e446d98         TCPv6    :::49154                       :::0                   LISTENING        880      svchost.exe
0x3e46e0c0         TCPv4    192.168.56.8:139               0.0.0.0:0              LISTENING        4        System
0x3e671988         TCPv4    0.0.0.0:445                    0.0.0.0:0              LISTENING        4        System
0x3e671988         TCPv6    ::445                          ::0                    LISTENING        4        System
0x3e72e530         TCPv4    0.0.0.0:135                    0.0.0.0:0              LISTENING        672      svchost.exe
0x3e72ee48         TCPv4    0.0.0.0:135                    0.0.0.0:0              LISTENING        672      svchost.exe
0x3e72ee48         TCPv6    :::135                         :::0                   LISTENING        672      svchost.exe
0x3e736388         TCPv4    0.0.0.0:49152                  0.0.0.0:0              LISTENING        340      wininit.exe
0x3e736d98         TCPv4    0.0.0.0:49152                  0.0.0.0:0              LISTENING        340      wininit.exe
0x3e736d98         TCPv6    :::49152                       :::0                   LISTENING        340      wininit.exe
0x3e79d528         TCPv4    0.0.0.0:49153                  0.0.0.0:0              LISTENING        724      svchost.exe
0x3e79d528         TCPv6    :::49153                       :::0                   LISTENING        724      svchost.exe
0x3e7a1618         TCPv4    0.0.0.0:49153                  0.0.0.0:0              LISTENING        724      svchost.exe
0x3e043b10         TCPv4    192.168.56.8:49177             192.168.56.5:80        CLOSED           -1
0x3ec6c820         UDPv4    127.0.0.1:51123                *:*                                     2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3f19a820         UDPv6    ::1:1900                       *:*                                     2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e8f0620         TCPv4    0.0.0.0:49156                  0.0.0.0:0              LISTENING        440      lsass.exe
0x3e9b3e50         TCPv4    0.0.0.0:49155                  0.0.0.0:0              LISTENING        432      services.exe
0x3e9b3e50         TCPv6    :::49155                       :::0                   LISTENING        432      services.exe
0x3ee1aa88         TCPv4    0.0.0.0:22                     0.0.0.0:0              LISTENING        2016     sshd.exe
0x3ee1bc18         TCPv4    0.0.0.0:22                     0.0.0.0:0              LISTENING        2016     sshd.exe
0x3ee1bc18         TCPv6    :::22                          :::0                   LISTENING        2016     sshd.exe
0x3fc0e008         UDPv4    127.0.0.1:1900                 *:*                                     2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3fc95d48         UDPv4    192.168.56.8:1900              *:*                                     2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3fcba008         TCPv4    192.168.56.8:49178             192.168.56.5:80        CLOSED           -1

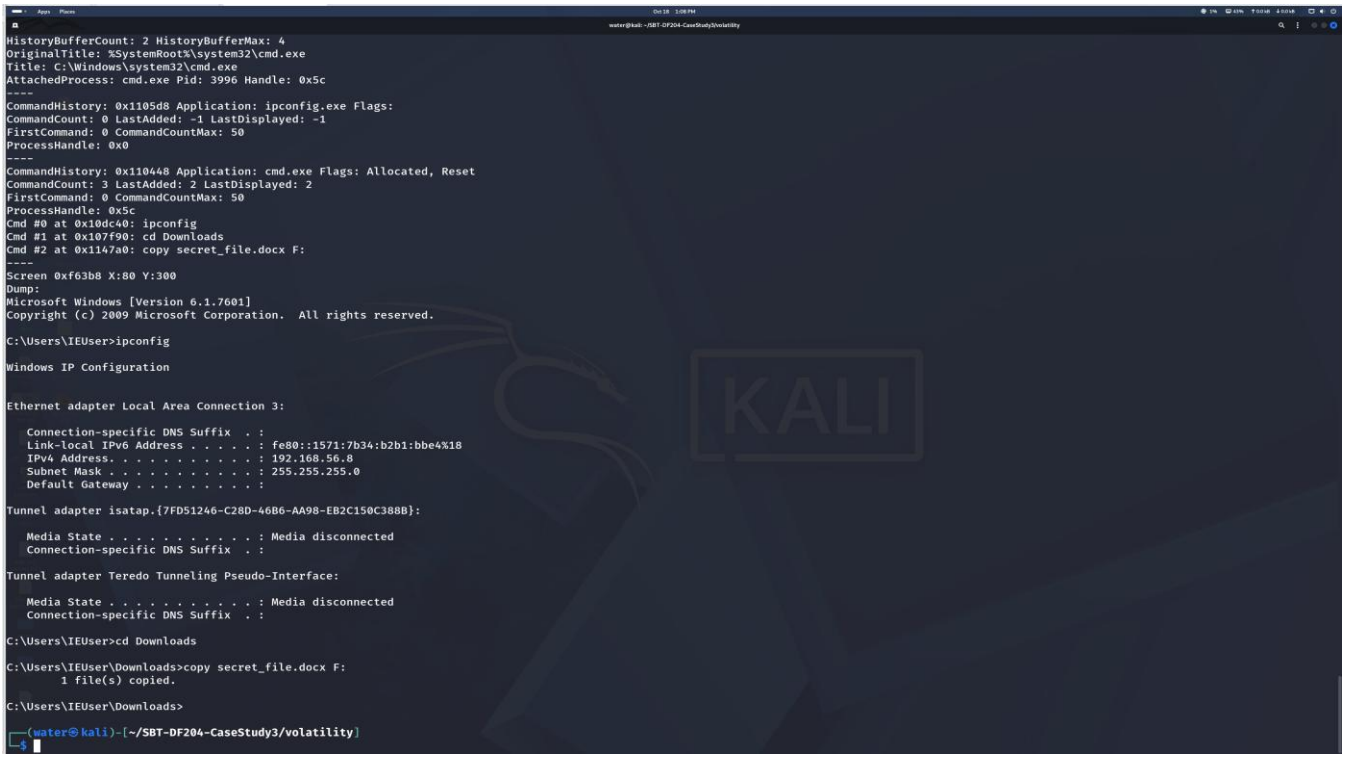
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$
```

Review command history



```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$ python2 ~/SBT-DF204-CaseStudy3/volatility/vol.py \
-f ~/SBT-DF204-CaseStudy3/working/memdumpWin7_working.mem \
--profile=Win7SP1x86 consoles \
2>&1 | tee ~/SBT-DF204-CaseStudy3/reports/consoles.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
**************************************************
ConsoleProcess: conhost.exe Pid: 2000
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
Title: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
AttachedProcess: sshd.exe Pid: 2016 Handle: 0x54
----
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
----
CommandHistory: 0x3207f0 Application: - Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x0
----
Screen 0x3360a0 X:80 Y:300
Dump:


**************************************************
ConsoleProcess: conhost.exe Pid: 3988
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
```



```
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: %SystemRoot%\system32\cmd.exe
Title: C:\Windows\system32\cmd.exe
AttachedProcess: cmd.exe Pid: 3996 Handle: 0x5c
----
CommandHistory: 0x1105d8 Application: ipconfig.exe Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x0
----
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 at 0x10dc40: ipconfig
Cmd #1 at 0x107f90: cd Downloads
Cmd #2 at 0x1147a0: copy secret_file.docx F:
----
Screen 0xf63b8 X:80 Y:300
Dump:
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation.  All rights reserved.

C:\Users\IEUser>ipconfig

Windows IP Configuration


Ethernet adapter Local Area Connection 3:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::1571:7b34:b2b1:bbe4%18
   IPv4 Address. . . . . . . . . . . : 192.168.56.8
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Tunnel adapter isatap.{7FD51246-C28D-46B6-AA98-EB2C150C388B}:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Tunnel adapter Teredo Tunneling Pseudo-Interface:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

C:\Users\IEUser>cd Downloads

C:\Users\IEUser\Downloads>copy secret_file.docx F:
        1 file(s) copied.

C:\Users\IEUser\Downloads>

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$
```

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$ python2 ~/SBT-DF204-CaseStudy3/volatility/vol.py \
-f ~/SBT-DF204-CaseStudy3/working/memdumpWin7_working.mem \
--profile=Win7SP1x86 cmdscan \
2>&1 | tee ~/SBT-DF204-CaseStudy3/reports/cmdscan.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
**************************************************
CommandProcess: conhost.exe Pid: 2000
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
Cmd #29 @ 0x3200c4: 3?4?2???2
Cmd #37 @ 0x3200c4: 3?4?2???2
**************************************************
CommandProcess: conhost.exe Pid: 3988
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:
Cmd #22 @ 0xff818488: ?
Cmd #25 @ 0xff818488: ?
Cmd #36 @ 0xe00c4: ?????
Cmd #37 @ 0x10cff0: ?????

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$
```

Locate USB and drive-mapping artefacts

Examine user accounts and registry hives

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$ python2 vol.py -f ~/SBT-DF204-CaseStudy3/working/memdumpWin7_working.mem --profile=Win7SP1x86 hivelist
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Virtual    Physical   Name
---------- ---------- ----
0x87c10370 0x27fd5370 [no name]
0x87c1c008 0x27fa3008 \REGISTRY\MACHINE\SYSTEM
0x87c459c8 0x27d8e9c8 \REGISTRY\MACHINE\HARDWARE
0x889c0430 0x1285a430 \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT
0x8b46d008 0x16d8c008 \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT
0x8b5d39b0 0x120049b0 \??\C:\Users\IEUser\ntuser.dat
0x97104008 0x1c45f008 \SystemRoot\System32\Config\SECURITY
0x9711d9c8 0x1e6a09c8 \SystemRoot\System32\Config\SOFTWARE
0x981f9008 0x1b446008 \SystemRoot\System32\Config\DEFAULT
0x98209518 0x1f3a7518 \SystemRoot\System32\Config\SAM
0x982d55c0 0x1e17c5c0 \REGISTRY\MACHINE\BCD00000000
0xa620b008 0x1997b008 \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa65269c8 0x17ad09c8 \??\C:\Users\sshd_server\ntuser.dat
0xa6539260 0x0f690260 \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa7d2a9c8 0x01c469c8 \??\C:\System Volume Information\Syscache.hve

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$
```

Examine USB devices and drive mappings

Checking USBSTOR with registry key

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$ python2 vol.py -f ~/SBT-DF204-CaseStudy3/working/memdumpWin7_working.mem --profile=Win7SP1x86 printkey -K "ControlSet001\Enum\USBSTOR"
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: USBSTOR (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:
  (S) Disk&Ven_General&Prod_UDisk&Rev_5.00

Values:

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$
```

Checking Device Classes

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$ python2 vol.py -f ~/SBT-DF204-CaseStudy3/working/memdumpWin7_working.mem --profile=Win7SP1x86 printkey -K "ControlSet001\Control\DeviceClasses"
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Legend: (S) = Stable   (V) = Volatile

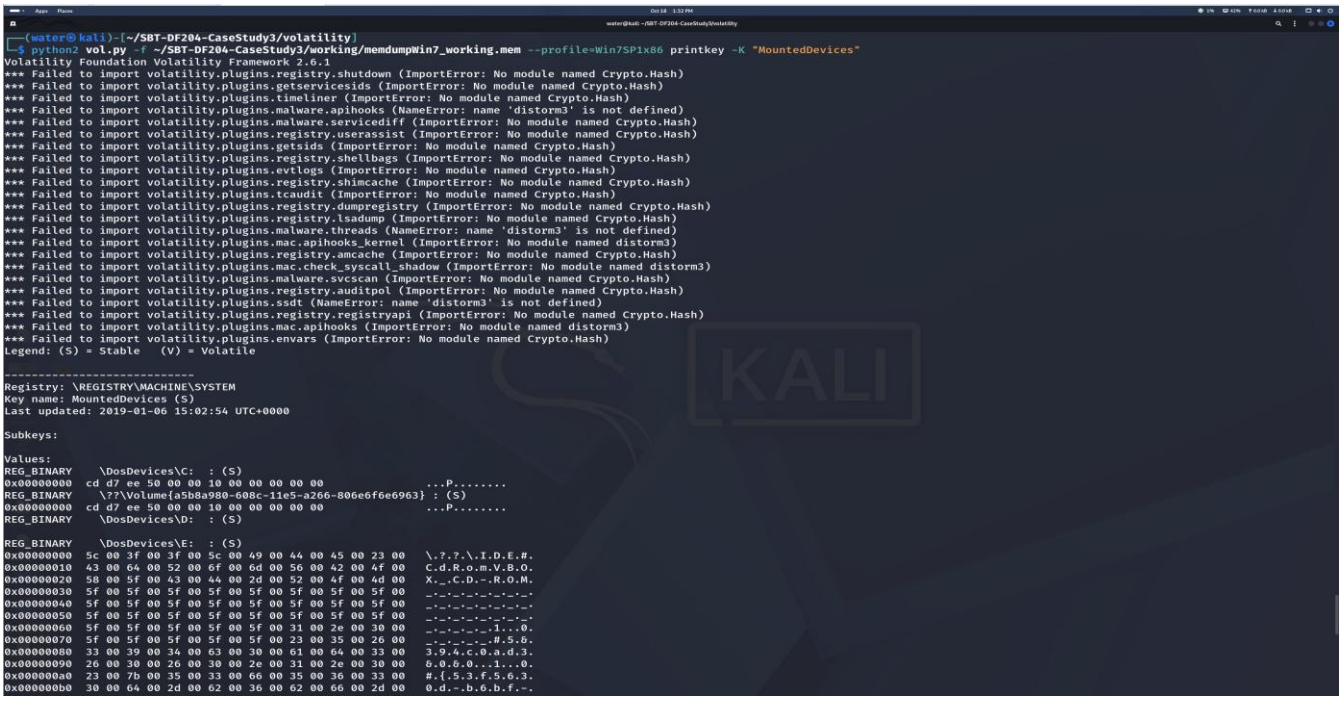
----------------------------
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: DeviceClasses (S)
Last updated: 2019-01-06 15:03:08 UTC+0000

Subkeys:
  (S) {0e0b6031-5213-4934-818b-38d90ced39db}
  (S) {10497b1b-ba51-44e5-8318-a65c837b6661}
  (S) {2accfe60-c130-11d2-b082-00a0c91efb8b}
  (S) {2e34d650-5819-42ca-84ae-d30803bae505}
  (S) {32412632-86cb-44a2-9b5c-50d1417354f5}
  (S) {3375baf4-9e15-4b30-b765-67acb10d607b}
  (S) {34d14be3-dee4-41c8-9ae7-6b174977c192}
  (S) {35fa2e29-ea23-4236-96ae-3a6ebacba440}
  (S) {3abf6f2d-71c4-462a-8a92-1e6861e6af27}
  (S) {4116f60b-25b3-4662-b732-99a6111edc0b}
  (S) {4afa3d53-74a7-11d0-be5e-00a0c9062857}
  (S) {4d1e55b2-f16f-11cf-88cb-001111000030}
  (S) {4d36e978-e325-11ce-bfc1-08002be10318}
  (S) {53f56307-b6bf-11d0-94f2-00a0c91efb8b}
  (S) {53f56308-b6bf-11d0-94f2-00a0c91efb8b}
  (S) {53f5630d-b6bf-11d0-94f2-00a0c91efb8b}
  (S) {53f5630e-b6bf-11d0-94f2-00a0c91efb8b}
  (S) {53f56311-b6bf-11d0-94f2-00a0c91efb8b}
  (S) {57164f39-9115-4e78-ab55-382f3bd5422d}
  (S) {5b45201d-f2f2-4f3b-85bb-30ff1f953599}
  (S) {6ac27878-a6fa-4155-ba85-f98f491d4f33}
```

```
  (S) {6ac27878-a6fa-4155-ba85-f98f491d4f33}
  (S) {72631e54-78a4-11d0-bcf7-00aa00b7b32a}
  (S) {866519b5-3f07-4c97-b7df-24c5d8a8ccb8}
  (S) {9527e630-d0ae-497b-adce-e80ab0175caf}
  (S) {97f76ef0-f883-11d0-af1f-0000f800845c}
  (S) {a9a0f4e7-5a45-4d96-b827-8a841e8c03e6}
  (S) {e849804e-c719-43d8-ac88-96b894c191e2}
  (S) {f33fdc04-d1ac-4e8e-9a30-19bbd4b108ae}

Values:
```

Checking Mounted Devices



```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy3/volatility]
└─$ python2 vol.py -f ~/SBT-DF204-CaseStudy3/working/memdumpWin7_working.mem --profile=Win7SP1x86 printkey -K "MountedDevices"
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (NameError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Legend: (S) = Stable   (V) = Volatile

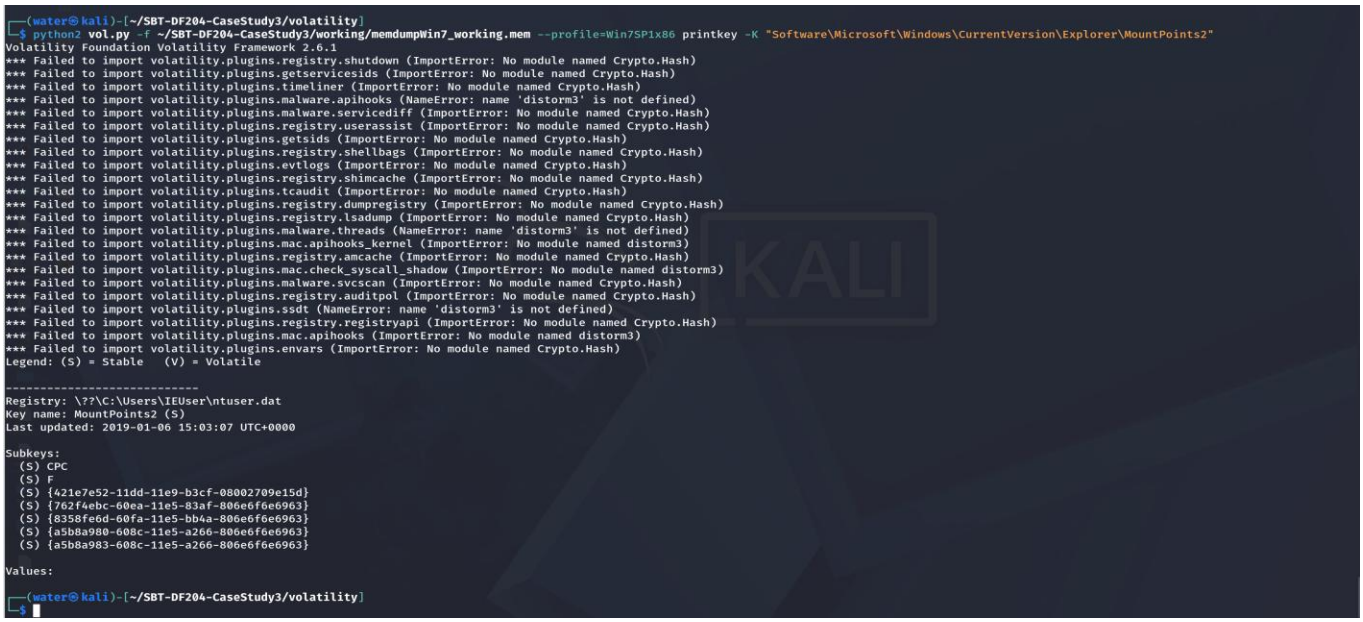
----------------------------
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: MountedDevices (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:

Values:
REG_BINARY    \DosDevices\C:  : (S)
0x00000000  cd d7 ee 50 00 00 10 00 00 00 00 00                  ...P........
REG_BINARY    \??\Volume{a5b8a980-608c-11e5-a266-806e6f6e6963} : (S)
0x00000000  cd d7 ee 50 00 00 10 00 00 00 00 00                  ...P........
REG_BINARY    \DosDevices\D:  : (S)

REG_BINARY    \DosDevices\E:  : (S)
0x00000000  5c 00 3f 00 3f 00 5c 00 49 00 44 00 45 00 23 00   \.?.?.\.I.D.E.#.
0x00000010  43 00 64 00 52 00 6f 00 6d 00 56 00 42 00 4f 00   C.d.R.o.m.V.B.O.
0x00000020  58 00 5f 00 43 00 44 00 2d 00 52 00 4f 00 4d 00   X._.C.D.-.R.O.M.
0x00000030  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00   _._._._._._._._.
0x00000040  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00   _._._._._._._._.
0x00000050  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00   _._._._._._._._.
0x00000060  5f 00 5f 00 5f 00 5f 00 31 00 2e 00 30 00         _._._._._1...0.
0x00000070  5f 00 5f 00 5f 00 5f 00 23 00 35 00 26 00         _._._._._.#.5.&.
0x00000080  33 00 39 00 34 00 63 00 30 00 61 00 64 00 33 00   3.9.4.c.0.a.d.3.
0x00000090  26 00 30 00 26 00 30 00 2e 00 31 00 2e 00 30 00   &.0.&.0...1...0.
0x000000a0  23 00 7b 00 35 00 33 00 66 00 35 00 36 00 33 00   #.{.5.3.f.5.6.3.
0x000000b0  30 00 64 00 2d 00 62 00 36 00 62 00 66 00 2d 00   0.d.-.b.6.b.f.-.
0x000000c0  31 00 31 00 64 00 30 00 2d 00 39 00 34 00 66 00   1.1.d.0.-.9.4.f.
0x000000d0  32 00 2d 00 30 00 30 00 61 00 30 00 63 00 39 00   2.-.0.0.a.0.c.9.
0x000000e0  31 00 65 00 66 00 62 00 38 00 62 00 7d 00         1.e.f.b.8.b.}.
REG_BINARY    \??\Volume{a5b8a985-608c-11e5-a266-806e6f6e6963} : (S)
0x00000000  5c 00 3f 00 3f 00 5c 00 46 00 44 00 43 00 23 00   \.?.?.\.F.D.C.#.
0x00000010  47 00 45 00 4e 00 45 00 52 00 49 00 43 00 5f 00   G.E.N.E.R.I.C._.
0x00000020  46 00 4c 00 4f 00 50 00 50 00 59 00 5f 00 44 00   F.L.O.P.P.Y._.D.
0x00000030  52 00 49 00 56 00 45 00 23 00 35 00 26 00 33 00   R.I.V.E.#.5.&.3.
0x00000040  61 00 61 00 33 00 39 00 34 00 37 00 65 00 26 00   a.a.3.9.4.7.e.&.
0x00000050  30 00 26 00 30 00 23 00 7b 00 35 00 33 00 66 00   0.&.0.#.{.5.3.f.
0x00000060  35 00 36 00 33 00 30 00 64 00 2d 00 62 00 36 00   5.6.3.0.d.-.b.6.
0x00000070  62 00 66 00 2d 00 31 00 31 00 64 00 30 00 2d 00   b.f.-.1.1.d.0.-.
0x00000080  39 00 34 00 66 00 32 00 2d 00 30 00 30 00 61 00   9.4.f.2.-.0.0.a.
0x00000090  30 00 63 00 39 00 31 00 65 00 66 00 62 00 38 00   0.c.9.1.e.f.b.8.
0x000000a0  62 00 7d 00                                       b.}.
REG_BINARY    \??\Volume{762f4ebc-60ea-11e5-83af-806e6f6e6963} : (S)
0x00000000  5c 00 3f 00 3f 00 5c 00 49 00 44 00 45 00 23 00   \.?.?.\.I.D.E.#.
0x00000010  43 00 64 00 52 00 6f 00 6d 00 56 00 42 00 4f 00   C.d.R.o.m.V.B.O.
0x00000020  58 00 5f 00 43 00 44 00 2d 00 52 00 4f 00 4d 00   X._.C.D.-.R.O.M.
0x00000030  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00   _._._._._._._._.
0x00000040  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00   _._._._._._._._.
0x00000050  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00   _._._._._._._._.
0x00000060  5f 00 5f 00 5f 00 5f 00 31 00 2e 00 30 00 30 00   _._._._.1...0.0.
0x00000070  5f 00 5f 00 5f 00 5f 00 23 00 35 00 26 00 30 00   _._._._.#.5.&.0.
0x00000080  33 00 39 00 34 00 63 00 30 00 61 00 64 00 33 00   3.9.4.c.0.a.d.3.
0x00000090  26 00 30 00 26 00 30 00 2e 00 30 00 2e 00 30 00   &.0.&.0...0...0.
0x000000a0  23 00 7b 00 35 00 33 00 66 00 35 00 36 00 33 00   #.{.5.3.f.5.6.3.
0x000000b0  30 00 64 00 2d 00 62 00 36 00 62 00 66 00 2d 00   0.d.-.b.6.b.f.-.
0x000000c0  31 00 31 00 64 00 30 00 2d 00 39 00 34 00 66 00   1.1.d.0.-.9.4.f.
0x000000d0  32 00 2d 00 30 00 30 00 61 00 30 00 63 00 39 00   2.-.0.0.a.0.c.9.
0x000000e0  31 00 65 00 66 00 62 00 38 00 62 00 7d 00         1.e.f.b.8.b.}.
REG_BINARY    \??\Volume{8358fe6d-60fa-11e5-bb4a-806e6f6e6963} : (S)
0x00000000  5c 00 3f 00 3f 00 5c 00 49 00 44 00 45 00 23 00   \.?.?.\.I.D.E.#.
0x00000010  43 00 64 00 52 00 6f 00 6d 00 56 00 42 00 4f 00   C.d.R.o.m.V.B.O.
0x00000020  58 00 5f 00 43 00 44 00 2d 00 52 00 4f 00 4d 00   X._.C.D.-.R.O.M.
0x00000030  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00   _._._._._._._._.
0x00000040  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00   _._._._._._._._.
0x00000050  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00   _._._._._._._._.
0x00000060  5f 00 5f 00 5f 00 5f 00 31 00 2e 00 30 00 30 00   _._._._.1...0.0.
0x00000070  5f 00 5f 00 5f 00 5f 00 23 00 35 00 26 00 30 00   _._._._.#.5.&.0.
0x00000080  33 00 39 00 34 00 63 00 30 00 61 00 64 00 33 00   3.9.4.c.0.a.d.3.
0x00000090  26 00 30 00 26 00 30 00 2e 00 30 00 2e 00 30 00   &.0.&.0...0...0.
0x000000a0  23 00 7b 00 35 00 33 00 66 00 35 00 36 00 33 00   #.{.5.3.f.5.6.3.
0x000000b0  30 00 64 00 2d 00 62 00 36 00 62 00 66 00 2d 00   0.d.-.b.6.b.f.-.
0x000000c0  31 00 31 00 64 00 30 00 2d 00 39 00 34 00 66 00   1.1.d.0.-.9.4.f.
0x000000d0  32 00 2d 00 30 00 30 00 61 00 30 00 63 00 39 00   2.-.0.0.a.0.c.9.
0x000000e0  31 00 65 00 66 00 62 00 38 00 62 00 7d 00         1.e.f.b.8.b.}.
REG_BINARY    \??\Volume{421e7e52-11dd-11e9-b3cf-08002709e15d} : (S)
0x00000000  5f 00 3f 00 3f 00 5f 00 55 00 53 00 42 00 53 00   _.?.?._.U.S.B.S.
0x00000010  54 00 4f 00 52 00 23 00 44 00 69 00 73 00 6b 00   T.O.R.#.D.i.s.k.
0x00000020  26 00 56 00 65 00 6e 00 5f 00 47 00 65 00 6e 00   &.V.e.n._.G.e.n.
0x00000030  65 00 72 00 61 00 6c 00 26 00 50 00 72 00 6f 00   e.r.a.l.&.P.r.o.
0x00000040  64 00 5f 00 55 00 44 00 69 00 73 00 6b 00 26 00   d._.U.D.i.s.k.&.
0x00000050  52 00 65 00 76 00 5f 00 35 00 2e 00 30 00 30 00   R.e.v._.5...0.0.
0x00000060  23 00 36 00 26 00 31 00 62 00 65 00 63 00 30 00   #.6.&.1.b.e.c.0.
0x00000070  66 00 34 00 38 00 26 00 30 00 26 00 5f 00 26 00   f.4.8.&.0.&._.&.
0x00000080  30 00 23 00 7b 00 35 00 33 00 66 00 35 00 36 00   0.#.{.5.3.f.5.6.
0x00000090  33 00 30 00 37 00 2d 00 62 00 36 00 62 00 66 00   3.0.7.-.b.6.b.f.
0x000000a0  2d 00 31 00 31 00 64 00 30 00 2d 00 39 00 34 00   -.1.1.d.0.-.9.4.
0x000000b0  66 00 32 00 2d 00 30 00 30 00 61 00 30 00 63 00   f.2.-.0.0.a.0.c.
0x000000c0  39 00 31 00 65 00 66 00 62 00 38 00 62 00 7d 00   9.1.e.f.b.8.b.}.
REG_BINARY    \DosDevices\F:  : (S)
0x00000000  5f 00 3f 00 3f 00 5f 00 55 00 53 00 42 00 53 00   _.?.?._.U.S.B.S.
0x00000010  54 00 4f 00 52 00 23 00 44 00 69 00 73 00 6b 00   T.O.R.#.D.i.s.k.
0x00000020  26 00 56 00 65 00 6e 00 5f 00 47 00 65 00 6e 00   &.V.e.n._.G.e.n.
0x00000030  65 00 72 00 61 00 6c 00 26 00 50 00 72 00 6f 00   e.r.a.l.&.P.r.o.
0x00000040  64 00 5f 00 55 00 44 00 69 00 73 00 6b 00 26 00   d._.U.D.i.s.k.&.
0x00000050  52 00 65 00 76 00 5f 00 35 00 2e 00 30 00 30 00   R.e.v._.5...0.0.
0x00000060  23 00 36 00 26 00 31 00 62 00 65 00 63 00 30 00   #.6.&.1.b.e.c.0.
0x00000070  66 00 34 00 38 00 26 00 30 00 26 00 5f 00 26 00   f.4.8.&.0.&._.&.
0x00000080  30 00 23 00 7b 00 35 00 33 00 66 00 35 00 36 00   0.#.{.5.3.f.5.6.
0x00000090  33 00 30 00 37 00 2d 00 62 00 36 00 62 00 66 00   3.0.7.-.b.6.b.f.
0x000000a0  2d 00 31 00 31 00 64 00 30 00 2d 00 39 00 34 00   -.1.1.d.0.-.9.4.
0x000000b0  66 00 32 00 2d 00 30 00 30 00 61 00 30 00 63 00   f.2.-.0.0.a.0.c.
0x000000c0  39 00 31 00 65 00 66 00 62 00 38 00 62 00 7d 00   9.1.e.f.b.8.b.}.
```

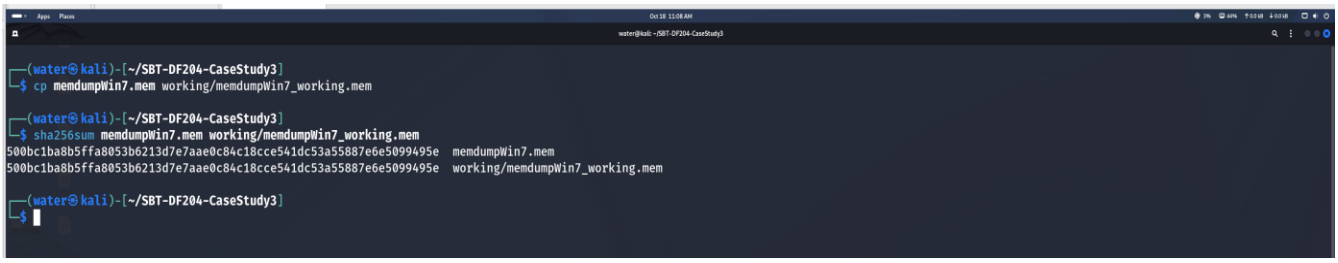
Checking Mount Points2



Below is the analysis and result to the eight questions asked on the case study and also relating to all the screenshot above

- **What evidence image did you acquire and how did you preserve it?,** Record the source, file name, file size, SHA-256 value, working-copy method, file-access precautions, and tools used.

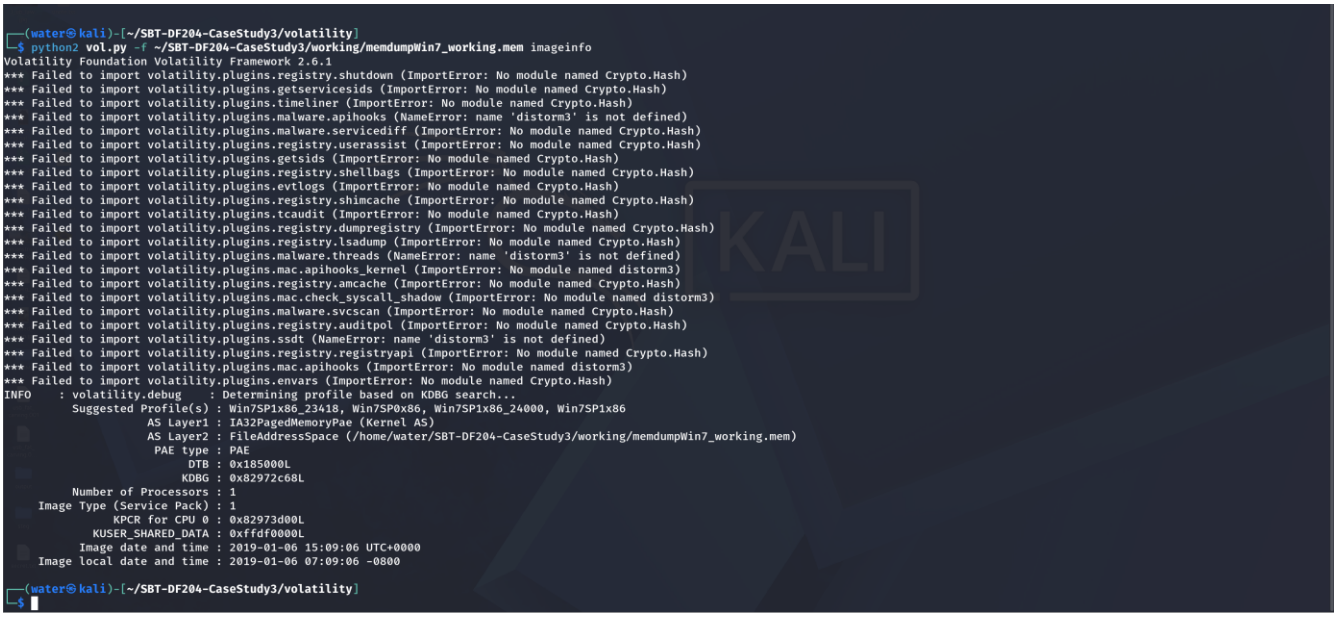
The evidence acquired was the Windows memory image named **memdumpWin7.mem**, supplied for the investigation of a suspected unauthorized file transfer. The original image was stored in the Case Study 3 folder, and its file details, including name, size, and permissions, were checked using Linux terminal commands.The SHA-256 hash recorded for the original image was **500bc1ba8b5ffa8053b6211d7e7aae0c84c18cce541dc53a55887e6e5099495e**.A working copy named working/memdumpWin7_working.mem was created using the cp command, and its SHA-256 hash was checked against the original to verify integrity.The evidence was preserved by keeping the original unchanged, analysing the working copy, and using Kali Linux, stat, sha256sum, cp, and Volatility 2 to support evidence handling and memory analysis.



sha256sum memdumpWin7.mem working/memdumpWin7_working.mem
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e memdumpWin7.mem
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e working/memdumpWin7_working.mem

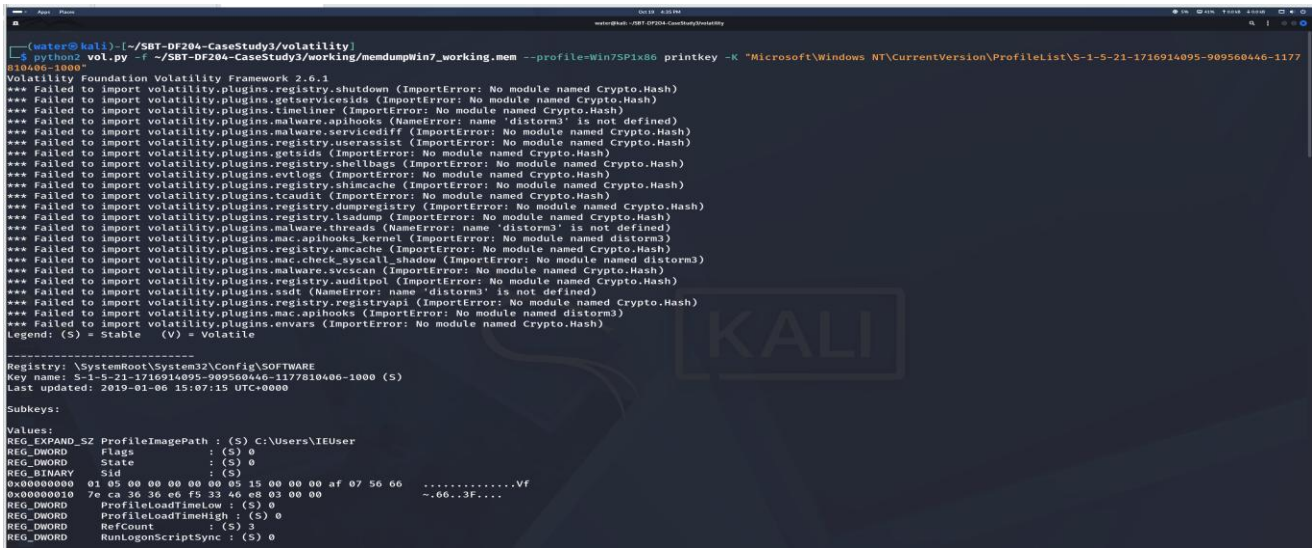
- **What operating-system profile and image time are supported?,** Provide imageinfo output, the candidate and selected profile, the image time, and the time zone. Explain how you validated the profile.

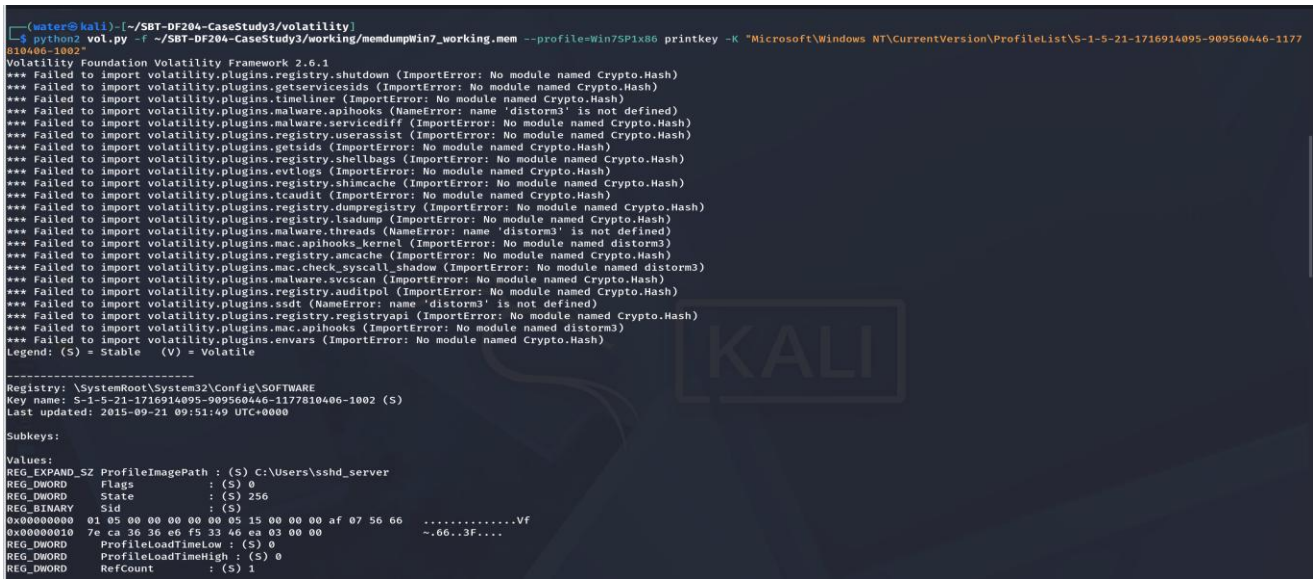
The imageinfo output from Volatility 2.6.1 identified the Windows 7 Service Pack 1 (SP1) 32-bit profile as a strong candidate for the memory image. The suggested profiles were Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, and Win7SP1x86. The recorded image time was **6 January 2019 at 15:09:06 UTC**, while the displayed local image time was **07:09:06 at UTC−08:00**.



- **What user and account artefacts are relevant?,** Identify the account or profile records examined, their SIDs or registry-hive paths, and the limit between profile evidence and proof of a person at the keyboard.

The relevant artefacts include Windows account records, user profiles, registry hives, and File Explorer activity records examined during the investigation. The Windows SOFTWARE registry hive contained the Microsoft\Windows NT\CurrentVersion\ProfileList key, which linked SID S-1-5-21-1716914095-909560446-1177810406-1000 to the profile C:\Users\IEUser, with a recorded update time of 6 January 2019 at 15:07:15 UTC. Another SID ending in 1002 was linked to C:\Users\sshd_server, with a recorded update time of 21 September 2015 at 09:51:49 UTC and a State value of 256. The SAM hive and individual NTUSER.DAT files are also relevant because they can provide account information and user-specific activity



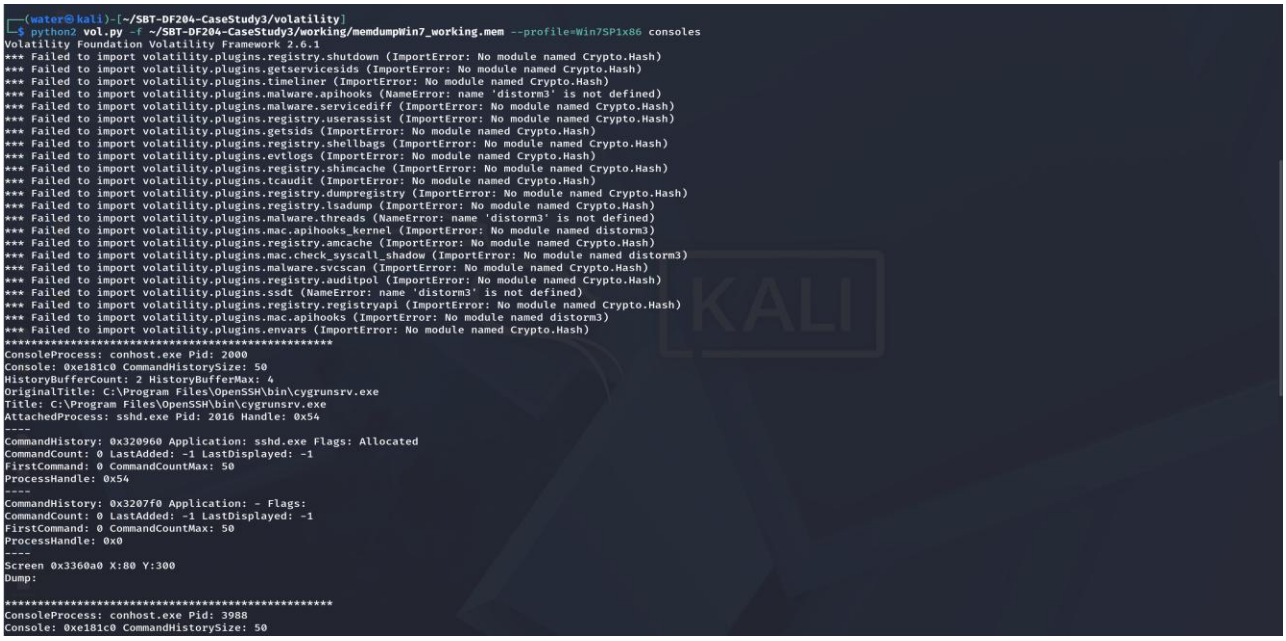


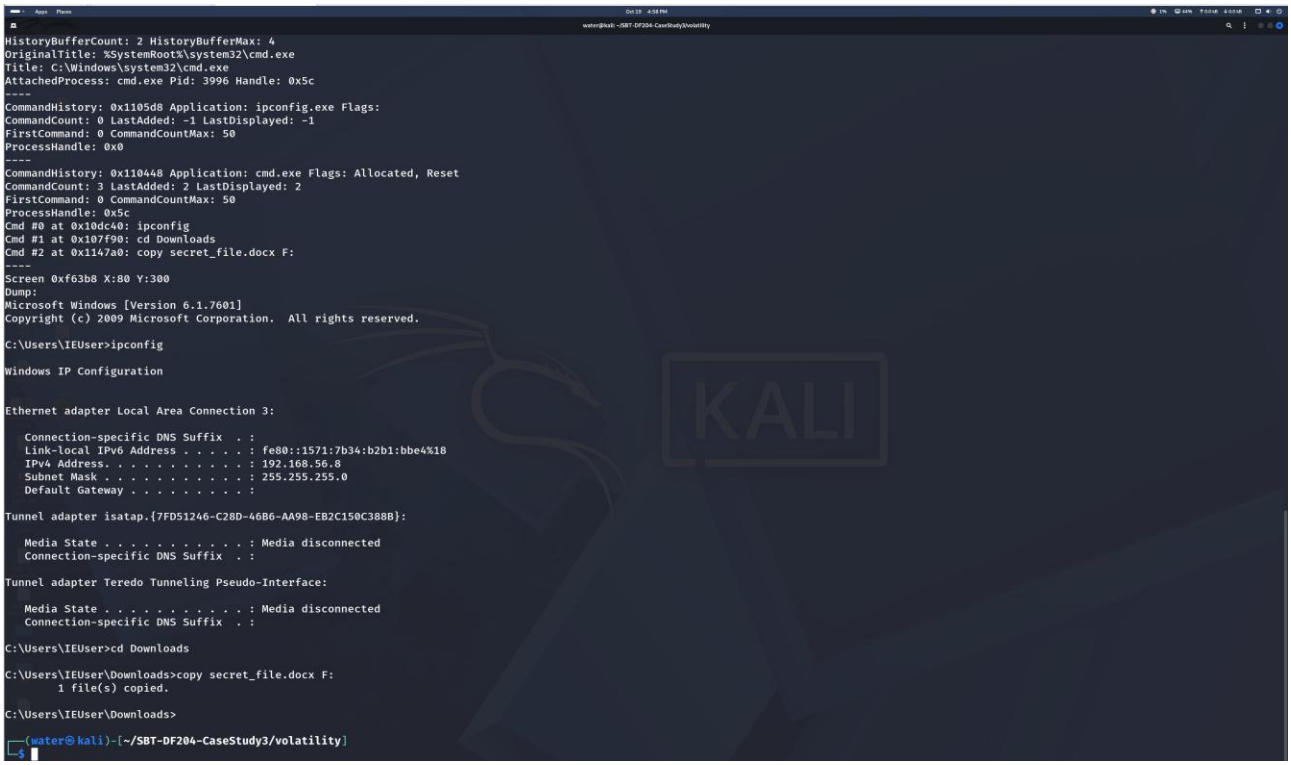
- **What process and network evidence relates to a potential download or transfer?** Identify relevant process IDs, process names, local and foreign endpoints, connection state, and memory strings or output records. Do not contact any endpoint.

The memory image was examined using Volatility to identify running processes and network connections that could be relevant to a potential file download or transfer. The pslist output identified chrome.exe with PIDs 2388, 1280, 2380, 2912, and 976; cmd.exe with PID 3996; sshd.exe with PID 2016; and FTK Imager.exe with PID 2596. The netscan output showed sshd.exe (PID 2016) listening on 0.0.0.0:22 and :::22, while a TCP connection from 192.168.56.8:49177 to 192.168.56.5:80 was recorded in the CLOSED state. Another connection from 192.168.56.8:49178 was also reported, but its complete destination and state was not verified from the original output; a UDP endpoint 192.168.56.8:56647 was associated with chrome.exe (PID 2388), with a recorded time of 6 January 2019 at 15:08:32 UTC. These findings indicate browser activity and network services in the captured system, but they do not independently establish that a file was downloaded or transferred.

- **What does command history show?** Use cmdscan or consoles output to identify relevant commands, the command process, path or drive references, and the timing or correlation evidence. Explain any limit on proving success.

The Volatility cmdscan and consoles outputs identified cmd.exe (PID 3996), attached to conhost.exe (PID 3988), as the process associated with the relevant command history. The recovered commands were ipconfig, cd Downloads, and copy secret_file.docx F:, and the console output confirmed the computer's IPv4 address as 192.168.56.8. Most importantly, the console displayed 1 file(s) copied, providing direct evidence that the copy command reported a successful transfer of secret_file.docx from C:\Users\IEUser\Downloads to the F: drive. The F: destination could be a removable drive, but its identity cannot be confirmed from command history alone; additional evidence, such as USB device registry records and file-system artefacts, is needed to establish the destination. The separate console associated with sshd.exe (PID 2016) contained no recoverable commands, so it does not independently establish SSH-based file transfer activity.





- **What removable-media evidence is present?** Use USBSTOR, DeviceClasses, MountedDevices, and MountPoints2 evidence as available to identify device, drive-letter, and user-profile relationships. State what each artefact does and does not establish.

The memory investigation recovered command history for cmd.exe (PID 3996), showing that the user navigated to C:\Users\IEUser\Downloads and executed copy secret_file.docx F:, after which the console reported 1 file(s) copied. This provides evidence of a successful copy operation to the F: drive, but the identity of the destination device was verified using Windows registry artefacts. The USBSTOR registry key under SYSTEM\ControlSet001\Enum\USBSTOR identify previously recognised USB storage devices and provide device identifiers, while DeviceClasses provide evidence of device-interface registrations. The SYSTEM\MountedDevices key help associate volume identifiers with drive letters, including F:, while NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 shows volume references associated with a particular user profile. The result from the case study confirm the copy command and the IEUser profile, but do not include the actual registry outputs needed to confirm a USB device model, serial number, F: volume mapping, or MountPoints2 association therefore, the removable media attribution remains unconfirmed. The strongest current evidence is that the Windows console reported a successful copy of secret_file.docx to F:. Confirming that F: was a particular USB storage device requires correlating the actual USBSTOR, DeviceClasses, MountedDevices, and MountPoints2 registry findings. Relevant profile: C:\Users\IEUser, associated with SID S-1-5-21-1716914095-909560446-1177810406-1000.

- **What conclusion can you defend?** Create a UTC timeline and state a cautious conclusion with a confidence level and at least three limitations or alternative explanations.

The memory investigation identified command history associated with cmd.exe (PID 3996), attached to conhost.exe (PID 3988), showing the commands ipconfig, cd Downloads, and copy secret_file.docx F:. The console output recorded the computer's IPv4 address as 192.168.56.8 and displayed 1 file(s) copied, supporting the conclusion that Windows reported a successful copy operation from C:\Users\IEUser\Downloads to the F: drive. The memory evidence also identified the IEUser profile and other running processes, including Chrome and the OpenSSH service, but their presence does not independently establish involvement in the file copy. Confidence is high that the recovered console recorded a copy command and reported success, but confidence is low that the destination was a specific removable USB device because the relevant registry evidence has not yet been confirmed. Limitations include the absence of verified USB device identifiers and drive mappings, uncertainty about the physical identity of the person operating the computer, and the possibility that the memory image does not contain all relevant historical activity. Therefore, the evidence supports a reported file copy to F:, but it does not conclusively establish the identity of the destination device or the person responsible.

UTC Timeline

Event	UTC timestamp	Evidence	Interpretation
User profile identified	Exact time not established	ProfileList registry evidence identifies C:\Users\IEUser.	Associates the profile directory with a Windows SID.
Network configuration displayed	Exact command time not established	ipconfig output shows 192.168.56.8.	Identifies the IPv4 address recorded in the console output.
Downloads directory accessed	Exact command time not established	cd Downloads in command history.	Indicates navigation to the Downloads directory.
File copy command issued	Exact command time not established	copy secret_file.docx F:.	Indicates an attempted copy to the F: drive.
Copy operation reported successful	Exact command time not established	Console output: 1 file(s) copied.	Supports a successful copy operation reported by Windows.
Memory image timestamp	2019-01-06 15:09:06 UTC	Previously reported case-study timestamp.	Provides a reference point for the memory evidence, not the exact time of the copy.

Timeline Template

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Browser or process evidence	2019-01-06 15:08:		Volatility netscan output	A UDP endpoint at 192.168.56.8:56647 was associated with	Indicates a browser-associated network endpoint. It does not independently prove

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
	32 UTC			chrome.exe, PID 2388.	that a file was downloaded or transferred.
Browser or process evidence	Not established		Volatility pslist output	Processes included chrome.exe (PIDs 2388, 1280, 2380, 2912, and 976), cmd.exe (PID 3996), sshd.exe (PID 2016), and FTK Imager.exe (PID 2596).	Identifies processes present in the memory image. Process presence alone does not prove involvement in the suspected file copy.
Browser or process evidence	Not established		Volatility netscan output	A TCP connection from 192.168.56.8:49177 to 192.168.56.5:80 was recorded in the CLOSED state.	Shows a recorded connection associated with the endpoints. It does not establish that a file was successfully transferred over that connection.
Browser or process evidence	Not established		Volatility netscan output	sshd.exe (PID 2016) was listening on 0.0.0.0:22 and :::22.	Indicates that the SSH service was listening. It does not prove that an SSH session or file transfer occurred.
Download-folder review	Not established		Command history from cmdscan	The command cd Downloads was recovered from the console associated with cmd.exe (PID 3996).	Indicates navigation to the Downloads directory. It does not establish which files were present in the directory at that time.
Download-folder review	Not established		Command history and console output	The command copy secret_file.docx F: was recovered.	Indicates an attempted copy from the current directory to the F: drive.

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Download-folder review	Not established		Windows console output recovered by Volatility	The console displayed C:\Users\IEUser\Downloads>copy secret_file.docx F: followed by 1 file(s) copied.	Strong evidence that Windows reported a successful copy operation. Independent verification of the destination file is not available in the supplied results.
Download-folder review	Not established		Windows console output from ipconfig	The console displayed IPv4 address 192.168.56.8 and subnet mask 255.255.255.0.	Records the network configuration displayed by the console. It does not identify the person operating the computer or establish a file transfer over the network.
Removable-device evidence	Not established		Windows SYSTEM hive, USBSTOR key	The actual USBSTOR device entries have not been supplied in the available case-study results.	A device model, serial number, or prior USB connection cannot yet be confirmed from the available evidence.
Removable-device evidence	Not established		Windows SYSTEM hive, ControlSet001\Control\DeviceClasses	The relevant device-interface registry output has not been supplied.	No specific device-interface registration can be confirmed from the available results.
Removable-device evidence	Not established		User NTUSER.DAT, Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2	The relevant MountPoints2 entries have not been supplied.	A connection between a specific mounted volume and the IEUser profile remains unverified.

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Removable-device evidence	Not established		Windows registry and file-system evidence	No verified destination-file listing, hash, or device-specific file-system artefact has been supplied.	The copied file and its destination have not been independently verified.
Drive mapping	Not established		Command history and console output	The destination specified in the copy command was F:.	Establishes the drive-letter reference used in the command, but not the physical identity of the destination volume.
Drive mapping	Not established		Windows SYSTEM\MountedDevices registry key	The actual MountedDevices output linking F: to a volume identifier has not been supplied.	A specific volume-to-drive mapping cannot yet be confirmed.
Drive mapping	Not established		Windows USBSTOR and MountedDevices registry evidence	The device identifiers and volume mapping have not been correlated in the available results.	It is not yet possible to conclude that F: was a particular USB storage device.
Command-history evidence	Not established		Volatility cmdscan output	conhost.exe PID 3988 contained a command history associated with cmd.exe PID 3996. The recovered commands were ipconfig, cd Downloads, and	Identifies the relevant command history and process relationship. The command-history output alone does not establish the exact execution time of each command.

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
				copy secret_file.docx F:.	
Command-history evidence	Not established		Volatility consoles output	The console associated with cmd.exe PID 3996 displayed the output of ipconfig.	Corroborates the recovered command and provides the network configuration displayed in the console.
Command-history evidence	Not established		Volatility consoles output	The console displayed 1 file(s) copied after the copy command.	Supports the conclusion that the copy operation was reported as successful, but does not identify the destination device.
Command-history evidence	Not established		Volatility cmdscan and consoles output	A separate console associated with sshd.exe PID 2016 contained no recoverable commands.	Does not independently establish SSH-based file transfer activity.
Supported conclusion	2019-01-06 15:09:06 UTC (memory image timestamp)		Memory image and Volatility analysis	The supplied case-study notes record this timestamp for the memory evidence.	Provides a reference point for the memory image, not the verified time of the copy command.
Supported conclusion	Not established		Combined command-history	Windows recorded the command copy secret_file.docx F:	High confidence that the recovered console reported a successful

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
			and console evidence	and displayed 1 file(s) copied.	copy operation; the physical destination remains unverified.
Supported conclusion	Not established		Combined process, network, profile, and command-history evidence	The IEUser profile and cmd.exe process are associated with the recovered command activity.	Supports an association with the Windows profile and process, but does not prove which individual was at the keyboard.
Supported conclusion	Not established		Registry evidence required for corroboration	USBSTOR, DeviceClasses, MountedDevices, and MountPoints2 findings are not available in the supplied results.	Confidence that the destination was removable media remains low until the registry evidence is examined and correlated.

Evidence log template

Evidence ID	Tool and plugin	Source record or path	Observed fact	Screenshot or appendix reference
E01	Volatility – windows.cmdscan	Console history associated with cmd.exe, PID 3996; conhost.exe, PID 3988	Recovered the commands ipconfig, cd Downloads, and copy secret_file.docx F:.	cmdscan output screenshot
E02	Volatility – windows.consoles	Console attached to cmd.exe, PID 3996; conhost.exe, PID 3988	The console displayed the IPv4 address 192.168.56.8 and subnet mask	Console output showing ipconfig

Evidence ID	Tool and plugin	Source record or path	Observed fact	Screenshot or appendix reference
			255.255.255.0 following ipconfig.	
E03	Volatility – windows.consoles	C:\Users\IEUser\Downloads; command copy secret_file.docx F:	The console displayed 1 file(s) copied, supporting a successful copy operation to the F: drive.	Console output showing the copy command and result
E04	Volatility – windows.pslist	Process record: cmd.exe, PID 3996	The process listing identified the command interpreter associated with the recovered command history.	pslist output screenshot
E05	Volatility – windows.consoles	Console attached to sshd.exe, PID 2016; conhost.exe, PID 2000	The console title referenced C:\Program Files\OpenSSH\bin\cygrunsrv.exe. No recoverable commands were displayed in the supplied output.	OpenSSH console output
E06	Volatility – windows.netscan	Network record: 192.168.56.8:49177 → 192.168.56.5:80; state CLOSED	The memory scan recorded a TCP connection involving the two endpoints in the CLOSED state. This does not prove a successful file transfer.	netscan output screenshot

2.7. Analysis and Findings

The Volatility analysis identified cmd.exe (PID 3996), attached to conhost.exe (PID 3988), with the commands ipconfig, cd Downloads, and copy secret_file.docx F:. The console output showed the computer's IPv4 address as 192.168.56.8 and confirmed that Windows reported 1 file(s) copied. The command history indicates that secret_file.docx was copied from C:\Users\IEUser\Downloads to the F: drive. The network scan also identified a TCP connection between 192.168.56.8:49177 and 192.168.56.5:80 in the CLOSED state, but this does not prove that the file was transferred over the network.
The available evidence supports a reported file copy, but the identity of the destination device and the person who performed the action remain unconfirmed.

2.8. Challenges and Solutions

One challenge was recovering reliable command history evidence from the Windows memory image. This was addressed by examining the Volatility windows.cmdscan and windows.consoles outputs.
Another challenge was distinguishing a successful copy command from proof that the destination was a removable USB device. It was resolve by examine the USBSTOR, Device Classes, Mounted Devices, and MountPoints2 registry artefacts and correlate their findings.
The third challenge was how to use ShellBags to identify material folder paths and recorded access or preference times. I was unable to work on this because the command was showing error



2.9. Conclusion

The investigation recovered command history showing that secret_file.docx was copied to the F: drive. The Windows console displayed 1 file(s) copied, providing strong evidence that the operating system reported the copy operation as successful.
Confidence is high in the recovered command and reported result, but the identity of the destination device remains unconfirmed. The evidence does not independently establish who was physically operating the computer when the command was entered.
Therefore, the defensible conclusion is that the captured Windows environment recorded a reported file copy, while attribution to a particular removable device or individual requires further evidence.

2.10. Recommendations

Assignment like this improved student knowledge and understanding on Forensic case giving the student the ability to think deep and do deep research before concluding on a case

2.11. References

Class materials Guide from Founder ICDFA on Basic Networking Skills for Digital Forensics, (Illegal File Transferring Memory Forensic)

https://onedrive.live.com/?id=%2Fpersonal%2Fa674908a5e2e2f2a%2FDocuments%2FDocuments%2Fdigital%2Dforensics%2Dlab%2FIllegal%5FFile%5FTransferring%5FMemory%5FForensics&listurl=%2Fpersonal%2Fa674908a5e2e2f2a%2FDocuments&e=tZtcsu&redeem=aHR0cHM6Ly8xZHJ2Lm1zL2YvYy9hNjc0OTA4YTVlMmUyZjJhL0lnREVtU0U5djdRb1JydjJXMlhFbDNMM0FVMjVHczBQd19WUTJQOS1ibVhVNWxRP2U9dFp0Y3N1&ga=1